

Institute of Engineering and Technology

Department of Computer Science & Engineering



Course Structure & Syllabus

B.Tech.

in

**Computer Science & Engineering-Artificial Intelligence & Data Science
(CSE-AI&DS)**

[Effective from the Session: 2026-2027]



INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of Computer Science & Engineering

Curriculum for Bachelor of Technology

CSE-AI&DS

Course Structure & Semester-wise Credit Distribution

Credit Distribution (Course Type) of B. Tech. (CSE- AI&DS) Program		
Sl. No.	Course Type	Credit
1	Basic Science Course (BSC)	14
2	Engineering Science Course (ESC)	08
3	Humanities & Social Science including Management Courses (HSMC)	09
4	Professional Core Course (PCC)	83
5	Professional Elective Course (PEC)	21
6	Project Work, Seminar , Internship in industry etc.	13
7	Ability Enhancement Course (AEC) offered by University	08
8	Skill Enhancement Course (SEC) offered by University	09
9	Open Elective Course (OEC)	12
10	ACITE-Mandatory Non-Credit (NC) Course(s)	
Total Credit		177



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Basic Science Courses (BSC)			
Sl. No.	Course Code	Course Title	Credits
1.	EPHY101	Engineering Physics	3+0
2.	EPHY151	Engineering Physics Lab	0+1
3.	ECHE101	Engineering Chemistry	3+0
4.	ECHE151	Engineering Chemistry Lab	0+1
5.	EMAT105	Mathematics for Data Science	3+0
6.	EMAT106	Mathematical Foundation of Machine Learning	3+0
Total			14
Engineering Science Course (ESC)			
1.	ECE101	Basic Electronics Engineering	3+0
2.	ECE151	Basic Electronics Engineering Lab	0+1
3.	ECE102	Basic Electrical Engineering	3+0
4.	ECE152	Basic Electrical Engineering Lab	0+1
Total			08
Humanities and Social Sciences including Management Courses (HSMC)			
1.	HSM101	Professional Communication	3+0
2.	HSM301	Organization Behavior	3+0
3.	HUV102	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	3+0
Total			9



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Professional Core Course (PCC)			
Sl. No.	Course Code	Course Title	Credit
1.	AID101	The C Language	3+0
2.	AID151	The C Language Lab	0+1
3.	AID102	Web Technology	3+0
4.	AID152	Web Technology Lab	0+1
5.	AID103	Python Programming	2+0
6.	AID153	Python Programming Lab	0+1
7.	AID104	Conversational Artificial Intelligence	3+0
8.	AID201	Data structure	3+0
9.	AID251	Data structure lab	0+1
10.	AID202	Database Management System	3+0
11.	AID252	Database Management System Lab	0+1
12.	AID203	Software Engineering	3+0
13.	AID204	Discrete Mathematics	3+0
14.	AID205	Theory of Computation	3+0
15.	AID206	Operating Systems	3+0
16.	AID254	Operating System Lab	0+1
17.	AID207	Data Science and Visualization Techniques	3+0
18.	AID255	Data Science and Visualization Techniques Lab	0+1
19.	AID208	Computer Network	3+0
20.	AID256	Computer Network Lab	0+1
21.	AID209	Probability and Statistics for Data Science	3+0
22.	AID301	Machine Learning	3+0
23.	AID351	Machine Learning Lab	0+1
24.	AID302	Design and Analysis of Algorithms	3+0
25.	AID352	Design and Analysis of Algorithm Lab	0+1
26.	AID303	Artificial Intelligence	3+0
27.	AID304	Data Mining and Data Warehousing	3+0
28.	AID305	Ethics for AI & Data Science	3+0
29.	AID306	Data Analytics and R programing	3+0
30.	AID307	Deep Learning & Neural Network	3+0
31.	AID353	Deep Learning & Neural Network Lab	0+1
32.	AID354	Advanced Java Programming Lab	0+1
33.	AID308	Computer Based Numerical and Statistical Techniques	3+0
34.	AID309	Natural Language Processing	3+0
35.	AID310	Computer Vision : Foundation and Applications	3+0
36.	AID311	Problem Solving Methods in Artificial Intelligence	3+0
Total			83



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Project Work, Seminar and Internship/Industrial Training in Industry etc.			
Sl. No.	Course Code	Course Title	Credit
1.	AIDIND401	Industrial Training	0+2
2.	AIDS401	Seminar	0+2
3.	AIDP401	Major Project I	0+3
4.	AIDP402	Major Project II	0+6
Total			13



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Professional Elective Course (PEC)			
Sl. No.	Semester	PEC-Elective	Credit
1	III	Elective-I	3+0
2.	IV	Elective-II	3+0
3.	V	Elective-I	3+0
4.	VI	Elective-II	3+0
5.	VII	Elective-I	3+0
6.	VII	Elective-I	3+0
7.	VIII	Elective-II	3+0
Total			21
Students have to select unique course(s) either from Elective-I or from Elective-II in the semester without repetition.			

Note: If a student has completed elective course(s) offered in higher semester(s) through SWAYAM, the corresponding credit(s) shall be duly recognized, and the student shall be exempted from appearing in the semester examination(s) for those course(s) in the respective semester(s).

Professional Open Elective Course (OEC)			
Sl. No.	Semester	OEC	Credit
1	VI	Open Elective-I	3+0
2.	VII	Open Elective-II	3+0
3.	VIII	Open Elective-I	3+0
4.	VIII	Open Elective-I	3+0
Total			12
Students have to select unique course(s) either from Elective-I or from Elective-II in the semester without repetition.			



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Department of Computer Science & Engineering

Professional Elective Courses (from SWAYAM)

Duration: 12 Weeks, Credit: 3

Elective-1 (Odd Semester)

Sl. No.	Course Code	Course Id	Course Name	SME Name	Institute
1.	EAID101	noc19_cs58	Computer Vision	Prof. Jayanta Mukhopadhyay	IITKGP
2.	EAID102	noc25_cs137	Programming with Generative AI	Prof. Vijay Kumar	IITs Bangalore
3.	EAID103	noc23-cs92	Artificial Intelligence : Search Methods For Problem Solving	Prof. Deepak Khemani	IITM
4.	EAID104	noc23-ge40	Fundamentals Of Artificial Intelligence	Prof. Shyamanta M. Hazarika	IITG
5.	EAID105	noc23-ee100	Applied Linear Algebra For Signal Processing, Data Analytics And Machine Learning	Prof. Aditya K. Jagannatham	IITK
6.	EAID106	noc26_cs61	Affective Computing	Prof. Jainendra Shukla, Prof. Abhinav Dhall	IIIT Delhi, IIT Ropar
7.	EAID107	noc19_cs82	Machine learning for Engineering and Science Applications	Prof. Balaji Srinivasan, Prof. Ganapathy Krishnamurthi	IIT Madras
8.	EAID108	noc20_cs39	Randomized Algorithm	Prof. Benny George K	IIT Guwahati

Note: In case of changes in the above subjects (Elective-1 (Even Semester) (from SWAYAM)), Dean/Director/ HoD will issue separate list of subjects.



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Elective-2 (Even Semester) (from SWAYAM)

Duration: 12 Weeks, Credit: 3

Sl. No.	Course Code	Course Id	Course Name	SME Name	Institute
1.	EAID201	<u>noc26-cs81</u>	Reinforcement Learning	Prof. Balaraman Ravindran	IIT Madras
2.	EAID202	<u>noc26-cs45</u>	Natural Language	Prof. Pawan Goyal	IIT Kharagpur
3.	EAID203	noc22_cs61	Learning Analytics Tools	Prof. Meenakshi D'souza	IIIT Bangalore
4.	EAID204	noc23-cs92	Deep Learning for Natural Language Processing	Prof. Deepak Khemani	IITM
5.	EAID205	noc23-ge40	Introduction to Large Language Models (LLMs)	Prof. Shyamanta M. Hazarika	IITG
6.	EAID206	noc23-ee100	Data Mining for Decision Making	Prof. Aditya K. Jagannatham	IITK
7.	EAID207	noc26_cs61	Programming, Data Structures And Algorithms Using Python	Prof. Jainendra Shukla, Prof. Abhinav Dhall	IIIT Delhi, IIT Ropar

Note: In case of changes in the above subjects (Elective-2 (Even Semester) (from SWAYAM)), Dean/Director/ HoD will issue separate list of subjects.



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ACITE- Mandatory Non-Credit (NC) Courses

Sl. No.	Course Code	Course Title	Credits
1.	----- --	Induction Program (3-weeks)	----- --

AICTE Mandatory 3-Week Student Induction Program

The Essence and Details of Induction program can also be understood from the 'Detailed Guide on Student Induction program', as available on AICTE Portal.

(Link: [https://www.aicte-india.org/sites/default/files/Model Curriculum/UG-1/ug-vol1.pdf](https://www.aicte-india.org/sites/default/files/Model%20Curriculum/UG-1/ug-vol1.pdf)).

Induction Program (mandatory)	Three-week duration
Induction Program for students to be offered right at the start of the first year.	Physical activity
	Creative Arts
	Universal Human Values-I
	Literary
	Proficiency Modules
	Lectures by Eminent People
	Visits to local Areas
	Familiarization to Dept./Branch & Innovations



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Open Elective I (from SWAYAM)

Sl. No.	Course Code	Course Id	Course Name	SME Name	Institute
1.	AIDOE01	noc19_cs58	Python for Data Science	Prof. Jayanta Mukhopadhyay	IITKGP
2.	AIDOE02	noc25_cs137	Introduction to Data Analytics	Prof. Vijay Kumar	IITs Bangalore
3.	AIDOE03	noc25-ag06	Machine Learning for Soil and Crop Management	Prof. Somsubhra Chakraborty	IIT Kharagpur
4.	AIDOE04	noc22_cs61	Data Analytics with Python	Prof. Meenakshi D'souza	IIIT Bangalore
5.	AIDOE05	noc26-cs02	Mathematical Foundations of Machine Learning	Prof. Prathosh AP	IISc Bangalore
6.	AIDOE06	noc25-ag10	Soil Science and Technology	Prof. Somsubhra Chakraborty	IIT Kharagpur
7.	AIDOE07	noc25-ag11	Traction Engineering	Prof. Hifjur Raheman	IIT Kharagpur
8.	AIDOE08	noc25-mg17	Data Analysis and Decision Making - I	Prof. Raghu Nandan Sengupta	IIT Kanpur
9.	AIDOE09	noc25-mg06	AI in Marketing	Prof. Zillur Rahman	IIT Roorkee
10.	AIDOE10	noc25-ge24	Non-conventional energy Resources	Prof. Prathap Haridoss	IIT Madras
11.	AIDOE11	noc25-mg66	Business Forecasting	Prof. Pankaj Dutta	IIT Bombay
12.	AIDOE12	noc25-mg38	Leadership and Team Effectiveness	Prof. Santosh Rangnekar	IIT Roorkee
13.	AIDOE13	noc25-mg08	Artificial Intelligence (AI) for Investments	Prof. Abhinava Tripathi	IIT Kanpur
14.	AIDOE14	noc25-hs68	Psychology of Stress, Health and Well-Being	Prof. Dilwar Hussain	IIT Guwahati
15.	AIDOE15	noc25-hs67	Psychology of Personality and Individual Differences: Theory and Applications	Prof. Dilwar Hussain	IIT Guwahati
16.	AIDOE16	noc25-hs72	Soft Skill Development	Prof. Priyadarshi Patnaik Prof. V. N. Giri Prof. D. Suar	IIT Kharagpur



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Open Elective II (from SWAYAM)

Sl. No.	Course code	Course Id	Course Name	SME Name	Institute
1.	AIDOE17	noc23_cs127	Cyber Security and Privacy	Prof. Saji K Mathew	IIT Madras
2.	AIDOE18	noc22_cs83	Applied Accelerated Artificial Intelligence	Prof. Satyajit Das	IIT Palakkad
3.	AIDOE19	noc22_cs53	Introduction to Internet of Things	Prof. Sudip Misra	IIT Kharagpur
4.	AIDOE20	noc26_cs06	Quantum Computing: Algorithm and Limitations Through the Query Model	Prof. Rajat Mittal	IIT Kanpur
5.	AIDOE21	nou26_cs13	Data science and Big Data	Prof. Sandeep Singh Rawat	IGNOU
6.	AIDOE22	noc23-hs131	Science, Technology and Society	Prof. Sambit Mallick	IITG
7.	AIDOE23	noc22_cs44	Blockchain and its Application	By Prof. Sandip Chakraborty, Prof. Shamik Sural	IIT Kharagpur
8.	AIDOE24	noc23-hs145	Cloud Computing	Prof. Soumya Kanti Ghosh	IITK
9.	AIDOE25	noc23-hs151	Public Speaking	Prof. Binod Mishra	IITR
10.	AIDOE26	noc23-hs155	Environmental Science	Prof. Samik Chowdhury & Prof. Sudha Goel	IITKGP
11.	AIDOE27	noc25_cs131	Big Data Computing	By Prof. Rajiv Misra	IIT Patna
12.	AIDOE28	noc24_cs86	Distributed Optimization and Machine Learning	By Prof. Mayank Baranwal	IIT Bombay
13.	AIDOE29	noc23-hs143	Educational Leadership	Prof. Atasi Mohanty	IITK
14.	AIDOE30	noc23-ma72	Non-Parametric	Prof. Niladri	IITD



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			Statistical Inference	Chatterjee	
15.	AIDOE31	noc23-mg122	Introduction to Marketing Essentials	Prof. Zillur Rahman	IITR
16.	AIDOE32	noc23-ec06	Principles of Economics	Prof. Sabuj Kumar Mandal	IITM



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Credit Distribution (Semester wise) (Minimum Common Program)

S.No.	Semester Number	Credit
1	Semester I	24
2	Semester II	25
3	Semester III	22
4	Semester IV	23
5	Semester V	26
6	Semester VI	26
7	Semester VII	16
8	Semester VIII	15
Total Credit		177

Note: Students, who have joined the job/internship/ summer training, can earn credits from MOOCS courses in the respective semesters as per course structure.



Course Structure



INSTITUTE OF ENGINEERING AND TECHNOLOGY
Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur

Course: CSE-AI&DS
Semester-I (First Year) Curriculum

Sl. No.	Type of Course	Course Code	Course Title	Credit
1.	Basic Science Course	ECHE101	Engineering Chemistry	3+0
2.	Basic Science Course	ECHE151	Engineering Chemistry Lab	0+1
3.	Basic Science Course	EMAT105	Mathematics for Data Science	3+0
4.	Engineering Science Course	AID101	The C Language	3+0
5.	Engineering Science Course	AID151	The C Language Lab	0+1
6.	Engineering Science Course	ECE101	Basic Electronics Engineering	3+0
7.	Engineering Science Course	ECE151	Basic Electronics Engineering Lab	0+1
8.	Professional Core Course	AID102	Web Technologies	3+0
9.	Professional Core Course	AID152	Web Technologies Lab	0+1
10.	SEC Course	SEC-x	SEC Course	03
11.	AEC Course	AEC-x	AEC Course	02
12.	Induction Program	-----		-----
Total Credit				24

Semester-II (First Year) Curriculum

Sl. No.	Type of Course	Course Code	Course Title	Credit
1.	Basic Science Course	EPHY101	Engineering Physics	3+0
2.	Basic Science Course	EPHY151	Engineering Physics Lab	0+1
3.	Basic Science Course	EMAT106	Mathematical Foundation of Machine Learning	3+0
4.	Humanities and Social Sciences including Management Courses	HSM101	Professional Communication	3+0
5.	Engineering Science Course	ECE101	Basic Electrical Engineering	3+0
6.	Engineering Science Course	ECE151	Basic Electrical Engineering Lab	0+1
7.	Professional Core Course	AID103	Python Programming	2+0
8.	Professional Core Course	AID153	Python Programming Lab	0+1
9.	Professional Core Course	AID104	Conversational Artificial Intelligence	3+0
10.	SEC Course	SEC-x	SEC Course	03
11.	AEC Course	AEC-x	AEC Course	02
Total Credit				25



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Deen Dayal Upadhyaya Gorakhpur University, Gorakhpur

Course: CSE-AI&DS
Semester-III (Second Year) Curriculum

Sl. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	AID201	Data Structure	3+0
2.	Professional Core Course	AID202	Database Management System	3+0
3.	Professional Core Course	AID203	Software Engineering	3+0
4.	Professional Core Course	AID204	Discrete Mathematics	3+0
5.	Professional Core Course	AID251	Data Structure Lab	0+1
6.	Professional Core Course	AID252	Database Management System Lab	0+1
7.	Professional Elective Course		Elective-I	3+0
8.	SEC Course	SEC-x	SEC Course	03
9.	AEC Course	AEC-x	AEC Course	02
Total Credit				22

Semester-IV (Second Year) Curriculum

Sl. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	AID205	Theory of Computation	3+0
2.	Professional Core Course	AID206	Operating Systems	3+0
4.	Professional Core Course	AID207	Data science and Visualization Techniques	3+0
5.	Professional Core Course	AID208	Computer Network	3+0
6.	Professional Core Course	AID209	Probability and Statistics for Data Science	3+0
7.	Professional Core Course	AID254	Operating System Lab	0+1
8.	Professional Core Course	AID255	Data Science and Visualization Techniques Lab	0+1
9.	Professional Core Course	AID256	Computer Network Lab	0+1
11.	Professional Elective Course		Elective-II	3+0
12.	AEC Course	AEC-x	AEC Course	02
Total Credit				23



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Course: CSE-AI&DS
Semester-V (Third Year) Curriculum

Sl. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	AID301	Machine Learning	3+0
2.	Professional Core Course	AID302	Design and Analysis of Algorithms	3+0
3.	Professional Core Course	AID303	Artificial Intelligence	3+0
4.	Professional Core Course	AID304	Data Mining and Data Warehousing	3+0
5.	Professional Core Course	AID305	Ethics for AI & Data Science	3+0
6.	Professional Core Course	AID306	Data Analytics using R Programming	3+0
7.	Professional Elective Course		Elective-I	3+0
8.	Humanities & Social Science including Management Courses	HSM301	Organization Behavior	3+0
9.	Professional Core Course	AID351	Machine Learning Lab	0+1
10.	Professional Core Course	AID352	Design and Analysis of Algorithm Lab	0+1
Total Credit				26

Semester-VI (Third Year) Curriculum

Sl. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	AID307	Deep Learning & Neural Network	3+0
2.	Professional Core Course	AID308	Computer Based Numerical and Statistical Techniques	3+0
3.	Professional Core Course	AID309	Natural Language Processing	3+0
4.	Professional Core Course	AID310	Computer Vision : Foundation and Applications	3+0
5.	Professional Core Course	AID311	Problem Solving Methods in Artificial Intelligence	3+0
6.	Professional Elective Course		Elective-II	3+0
7.	Open Elective Course		Open Elective-I	3+0
8.	Humanities & Social Science including Management Courses	HUV102	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	3+0
9.	Professional Core Course	AID353	Deep Learning & Neural Network Lab	0+1
10.	Professional Core Course	AID354	Advanced Java Programming Lab	0+1
Total Credit				26



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Course: CSE-AI&DS
Semester-VII (Fourth Year) Curriculum

Sl. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Elective Course		Elective-I	3+0
2.	Professional Elective Course		Elective-I	3+0
3.	Open Elective Course		Open Elective-II	3+0
4.	Project work, Seminar and internship in industry or elsewhere	AIDS401	Seminar	0+2
5.	Project work, Seminar and internship in industry or Elsewhere	AIDIND401	Industrial Training	0+2
6.	Project work, Seminar and internship in industry or Elsewhere	AIDP401	Major Project-I	0+3
Total Credit				16

Semester-VIII (Fourth Year) Curriculum

Sl. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Elective Course		Elective-II	3+0
2.	Open Elective Course		Open Elective-I	3+0
3.	Open Elective Course		Open Elective-I	3+0
4.	Project work, Seminar and internship in industry or Elsewhere	AIDP402	Major Project-II	0+6
Total Credit				15

Syllabus

Syllabus of B.Tech. Computer Science & Engineering-Artificial Intelligence & Data Science

The C Language

Course Code: AID101

Credit: 3+0

Course Objective: Students will be able to develop logics which will help them to create programs, applications in C. Also, by learning the basic programming constructs they can easily switch over to any other language in future.

Syllabus

UNIT 1: Introduction to Programming: Introduction to components of a computer system, Algorithm: Representation of Algorithm, Flowchart, Pseudo code with examples, from algorithms to programs, source code. Programming Basics: Structure of C program: writing and executing the first C program, Syntax and logical errors in compilation, object and executable code. Components of C language: Standard I/O in C, Fundamental data types, Variables and memory locations, Storage classes. Arithmetic expressions & Conditional Branching: Arithmetic expressions and precedence: Operators and expression using numeric and relational operators, mixed operands, type conversion, logical operators, bit operations, assignment operator, operator precedence and associativity. Conditional Branching: Applying if and switch statements, nesting if and else, use of break and default with switch case.

UNIT 2: Loops & Functions: Iteration and loops: use of while, do while and for loops, multiple loop variables, use of break and continue statements. Functions: Introduction, types of functions, functions with array, passing parameters to functions, call by value, call by reference, recursive.

UNIT 3: Arrays & Basic Algorithms: Array notation and representation, manipulating array elements, using multi-dimensional arrays. Character arrays and strings, Structure, union, enumerated data types, Array of structures, passing arrays to functions. Basic Algorithms: Searching & Basic Sorting Algorithms (Bubble, Insertion and Selection), Finding roots of equations, Notion of order of complexity.

UNIT 4: Pointer & File Handling: Pointers: Introduction, declaration, applications, Introduction to dynamic memory allocation (malloc, calloc, realloc, free), Use of pointers in self-referential structures, notion of linked list (no implementation) File handling: File I/O functions, Standard C pre-processors, defining and calling macros, command-line arguments.

Course Outcomes (COs): At the end of this course students will be able to:

CO1: To develop simple algorithms for arithmetic and logical problems.

CO2: To translate the algorithms to programs & execution (in C language).

CO3: To implement conditional branching, iteration and recursion.

Syllabus of B.Tech. Computer Science & Engineering-Artificial Intelligence & Data Science

C04: To decompose a problem into functions and synthesize a complete program using divide and conquer approach.

C05: To use arrays, pointers and structures to develop algorithms and programs.

Textbooks:

1. The C programming by Kernighan Brain W. and Ritchie Dennis M., Pearson Education.
2. Computer Basics and C Programming by V. Rajaraman, PHI Learning pvt. Limited, 2015.
3. Computer Concepts and Programming in C, E Balaguruswami, McGraw Hill
4. Computer Science- A Structured Programming Approach Using C, by Behrouz A. Forouzan, Richard F. Gilberg, Thomson, Third Edition, Cengage Learning - 2007.
5. Let Us C By Yashwant P. Kanetkar, BPB Publication.
6. Problem Solving and Program Design in C, by Jeri R. Hanly, Elliot B. Koffman, Pearson Addison-Wesley, 2006.
7. Programming in C by Kochan Stephen G. Pearson Education – 2015.
8. Computer Concepts and Programming by Anami, Angadi and Manvi, PHI Publication.

Syllabus of B.Tech. Computer Science & Engineering-Artificial Intelligence & Data Science

The C Language Lab

Course Code: AID151

Credit: 0+1

Experiments

1. Write programs to print statements in sequential order using simple printf, scanf input/output functions.
2. Write programs to implement if-else conditions (simple as well as nested) on suitable problems.
3. Write a program to implement switch-case conditional logic on suitable examples.
4. Write programs to implement for, while and do-while loop control statements on suitable problems.
5. Write programs to implement 1D & 2D array concepts on suitable problems such as sorting of elements, searching of element, matrix addition, subtraction, multiplication etc.
6. Write programs to implement string related concepts such as sorting of a string, finding its length, reversing, concatenation, comparing two strings etc.
7. Write programs to implement concepts of user defined functions (call by value, call by reference, recursive calling etc.) on suitable examples.
8. Write programs to implement concepts of pointer.
9. Write programs to implement the concept of structure and union.
10. Write programs to implement dynamic memory allocation functions (calloc, malloc, free, realloc)
11. Write programs to implement file handling concepts such as reading from a file, writing to a file using file related functions (fclose, fopen, sscanf, sprintf, fread, fwrite,getc, putc, getw, putw etc.)

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

Syllabus of B.Tech. Computer Science & Engineering-Artificial Intelligence & Data Science

Web Technologies

Course Code: AID102

Credit: 3+0

Course Objective: Students should be able to identify common web design errors, create and edit web pages using appropriate tools, apply HTML effectively, and demonstrate a working understanding of CSS, XML, JavaScript, JSP, ASP.NET.

Syllabus

UNIT 1: Introduction to WWW - World Wide Web, WWW Architecture, Web Search Engines, Web Crawling, Web Indexing, Web Searching, Search Engines Optimization and Limitations, Web Mining: Web Content Mining, Web Structure Mining, Web Usage Mining

UNIT 2: Markup Language Basics: SGML, HTML, CSS And XML

SGML: Standard Generalized Markup Language (SGML) -Structures, Elements, Content Models, DTD, Attributes Entities.

HTML: Designing Web Pages with HTML-Use of Tags, Hyperlinks, URLs, Tables, Text Formatting, Graphics & Multimedia, Imagemap, Frames and Forms in Web Pages.

CSS: Use of Cascading Style Sheet in Web Pages.

XML: Extensible Markup Language (XML): Introduction using User-Defined Tags in Web Pages, Displaying XML Contents, XML Dtds, Use of XSL

UNIT 3: Client-Side Scripting using JAVA Script

JAVA script Overview; Constants, Variables, Operators, Expressions & Statements; User-Defined & Built-in Functions; Client-Side Form Validation; Using Properties and Methods of Built-in Objects.

UNIT 4: Server-Side Scripting Using JSP, ASP.NET

JSP: Introduction to JSP, JSP Architecture, JSP Directives, JSP Scripting Elements, Default Objects in JSP, JSP Actions, JSP with Beans and JSP with Database, Error Handling in JSP, Session Tracking Techniques in JSP, Introduction to Custom Tags.

ASP.NET: ASP.Net Coding Modules, ASP.NET Page Directives, Page Events and Page Life Cycle, Postback and Crosspage Posting ASP.NET Server Controls, HTML Controls, Validation Controls, Building Databases.

Course Outcomes (COs): After completing this course, students will be able to:

CO1: Understand the fundamentals of web technologies including client-server architecture, HTTP/HTTPS, and web services relevant to AI-based applications.

CO2: Design and develop responsive web interfaces using HTML, CSS, and JavaScript for data-driven applications.

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C03: Implement dynamic web applications and integrate them with APIs to fetch, display, and process real-time data.

C04: Develop and validate web forms for efficient data collection, preprocessing, and user interaction in AI systems.

C05: Connect web applications with backend systems and databases to store, retrieve, and manage large datasets.

C06: Apply basic concepts of authentication, security, and data privacy in web-based AI and data science applications.

C07: Build simple web-based dashboards and visualization tools for presenting data insights using modern frameworks/libraries.

Textbooks:

1. Uttam K. Roy, Web Technologies, 1/e, Oxford University Press, USA
2. M. Srinivasan, Web Technology: Theory and Practice, Pearson Education India
3. Deitel, Deitel and Nieto, Internet and Worldwide Web - How to Program, 5th Edition, PHI, 2011
4. Ralph Moseley & M. T. Savaliya, Developing Web Application- Second Edition, Wiley

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Web Technology Lab

Course Code: AID152

Credit: 0+1

Experiments

1. Create a HTML static web page which shows the use of different tags in that.
2. Insert an image and create a link such that clicking on the image takes the user to another page.
3. Prepare a sample code to illustrate three types of lists in HTML.
4. Use tables to provide layout to your HTML page describing your university infrastructure
5. Use frames such that the page is divided into 3 frames 20% on left to show contents of pages, 60% in center to show body of page, remaining on right to show remarks.
6. Create a simple form that will show all the INPUT METHODS available in HTML.
7. Create a sample code to illustrate the Embedded, External and Inline style sheets for your web page.
8. Write down simple JAVA Script using timeout such that the image will be changed after every 1 ms at a specified position.
9. Design a registration form and validate its field by using JAVA script.
10. Write an XML example of given tree that demonstrates the creation of user-designed tags and display it in a browser
11. Write a program in XML for creation of DTD which specifies a particular set of rules.
12. Create a bean student with attributes (first name, last name, age, class). In another JSP page display the bean values using <jsp:usebean>.
13. Write a program to use JDBC connectivity program for maintaining database by sending queries through JSP Page.
14. Use ad-rotator to change advertisements on client-side request (ASP.NET)
15. Implement Session tracking using user authentication in ASP.NET.

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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Python Programming

Course Code: AID103

Credit: 3+0

Course Objective: Students should be able to develop a strong understanding of fundamental concepts, enhance analytical and problem-solving skills for real-world applications, build a solid foundation for advanced study and research, apply knowledge creatively and practically, and acquire skills relevant to industry and professional requirements.

Syllabus

UNIT 1: Introduction to Python: Python Basics, Variables and Data Types, Basic Input and Output, Operators and Expressions, Control Flow: if, elif, else, Loops: for and while, Lists and Tuples, Dictionaries and Sets, Functions: Syntax and Arguments, Built-in Functions and Libraries.

UNIT 2: Advanced Python Concepts: File Handling: Reading and Writing Files, Exception Handling: try, except, finally, Comprehensions: List, Dictionary, and Set Comprehensions, Iterators and Generators, Decorators: Function Decorators, Class Decorators, Object-Oriented Programming (OOP): Classes and Objects, Inheritance and Polymorphism, Encapsulation and Abstraction, Magic Methods.

UNIT 3: Working with Data in Python: Working with CSV and JSON Data, Working with Excel Spreadsheet's, Data Visualization with Matplotlib, Data Analysis with Pandas: Series and Data Frames, Data Cleaning and Preprocessing, Grouping and Aggregation, Merging and Joining Data, Time Series Analysis, Introduction to NumPy.

UNIT 4: Python Standard Library and Modules: Overview of Python Standard Library, Commonly used modules (os, sys, datetime, math, random), Working with data (csv, json, sqlite3), Web scraping (requests, BeautifulSoup, scrapy), GUI programming (tkinter, PyQt, Kivy), Data compression and archiving (zipfile, tarfile), Working with images and audio (Pillow, pygame), Interacting with the operating system (subprocess, shutil), Email handling (smtplib, imaplib), Logging and debugging utilities (logging, pdb), Working with dates and times (datetime, time), Internationalization and localization (gettext, locale)

Course Outcomes (COs): After completing this course, students will be able to:

CO1: Understand the basic concepts of Python programming including syntax, data types, operators, and control structures.

CO2: Apply Python programming constructs such as functions, modules, and file handling to solve computational problems.

CO3: Implement object-oriented programming concepts (classes, objects, inheritance, polymorphism) using Python.

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C04: Utilize Python libraries and packages (such as NumPy, Pandas, Matplotlib) for data manipulation and analysis.

C05: Develop programs to process, analyze, and visualize data for data science applications.

C06: Implement basic algorithms and problem-solving techniques using Python for real-world scenarios.

C07: Build simple applications integrating Python with databases, APIs, or web frameworks for AI & DS tasks.

Textbooks:

1. "Effective Python: 90 Specific Ways to Write Better Python" by Brett Slatkin
2. "Fluent Python: Clear, Concise, and Effective Programming" by Luciano Ramalho
3. "Python Cookbook" by David Beazley and Brian K. Jones
4. "Learning Python" by Mark Lutz
5. "Python for Data Analysis" by Wes McKinney

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Python Programming Lab

Course Code: AID153

Credit: 0+1

Experiments

1. Write a Python program to perform basic input/output operations and demonstrate variables and data types.
2. Create a program to perform arithmetic, relational, and logical operations on user input.
3. Implement programs using if, elif, and else (e.g., grading system, largest of three numbers).
4. Write programs using for and while loops (e.g., factorial, Fibonacci series).
5. Implement operations on lists, tuples, dictionaries, and sets (e.g., searching, sorting, updating).
6. Create user-defined functions and use built-in functions (e.g., len(), sum(), max()).
7. Write a program to read from and write to text files.
8. Implement programs using try, except, and finally to handle runtime errors.
9. Create list, dictionary, and set comprehensions for various tasks.
10. Develop programs demonstrating classes, objects, inheritance, and polymorphism.
11. Implement custom iterators, generators, and function decorators.
12. Read, write, and manipulate CSV and JSON data files.
13. Perform data manipulation using Series and DataFrames.
14. Create plots (line, bar, histogram, scatter) using Matplotlib.
15. Perform data preprocessing (handling missing values) and basic NumPy operations.

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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Conversational Artificial Intelligence

Course Code: AID104

Credit: 3+0

Course Objective: The course aims to provide students with a fundamental understanding of conversational AI and its applications in real-world systems. It focuses on developing knowledge of natural language processing techniques required for building intelligent chatbots. Students will learn how to design, implement, and manage dialogue systems using modern tools and frameworks. The course also emphasizes developing practical skills to build and deploy conversational agents. Additionally, it introduces ethical considerations such as data privacy, bias, and responsible AI usage.

Syllabus

UNIT 1: Introduction to Conversational AI: Introduction to Conversational AI and Chatbots, Types of Chatbots: Rule-based vs AI-based, History and Evolution of Conversational Systems, Applications of Conversational AI (Healthcare, E-commerce, Banking, Education), Basics of Natural Language Processing (NLP), Text Preprocessing: Tokenization, Stop-word removal, Stemming, Lemmatization, Overview of Dialogue Systems Architecture, Introduction to Speech-based vs Text-based Systems.

UNIT 2: Natural Language Understanding (NLU): Syntax and Semantics in NLP, Intent Recognition and Entity Extraction, Named Entity Recognition (NER), Part-of-Speech (POS) Tagging, Text Classification Techniques, Word Embeddings (Word2Vec, GloVe), Introduction to Transformers and Attention Mechanism, Pre-trained Language Models (BERT, GPT basics).

Dialogue Management: Components of Dialogue Systems, Dialogue State Tracking, Rule-based Dialogue Management, Machine Learning-based Dialogue Systems,

UNIT 3: Reinforcement Learning: Reinforcement Learning for Dialogue Systems, Context Handling in Conversations, Multi-turn Conversations and Memory, Error Handling and Fallback Strategies.

Natural Language Generation (NLG) & Tools: Introduction to Natural Language Generation, Template-based vs AI-based Response Generation, Text Generation Techniques, Evaluation Metrics for Conversational Systems, Conversational AI Platforms and Tools: Dialogflow, Rasa, Microsoft Bot Framework, Building Chatbots using Rasa/Dialogflow, Integration with APIs and Databases, Voice Assistants and Speech Processing (ASR, TTS basics).

Unit 4: Advanced Applications of Conversational AI: Building End-to-End Chatbot Systems, Conversational AI with Deep Learning, Multilingual Chatbots and Translation Systems, Sentiment Analysis in Conversations, Ethics in Conversational AI (Bias, Privacy, Security), Deployment of Chatbots (Web, Mobile, Cloud), Integration with IoT and Smart Systems, Future Trends: Generative AI, LLMs, and Human-AI Interaction.

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Course Outcomes (COs): After completing this course, students will be able to:

C01: Understand the fundamentals of conversational AI, chatbot types, architectures, and NLP basics.

C02: Apply natural language understanding techniques such as intent classification, entity recognition, and text preprocessing.

C03: Design and implement dialogue management systems for handling multi-turn conversations.

C04: Develop conversational agents using tools like Rasa, Dialogflow, or similar frameworks and integrate with APIs.

C05: Build and deploy end-to-end conversational AI applications using machine learning and deep learning techniques.

C06: Evaluate conversational systems and address ethical issues such as bias, privacy, and security.

Textbooks:

1. Hands-on chatbot with Google Dialogflow, Loonycorn, O'Reilly, Packt publishing
2. Hands-on chatbots and conversational UI development, Srini Janarthnam, Packt publishing

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Data Structures

Course Code: AID201

Credit: 3+0

Course Objective: Students should be able to understand fundamental data structures and their implementations, recognize their role in efficient programming, and apply appropriate data structures to solve computational problems effectively.

Syllabus

UNIT 1: Introduction: Basic Terminology, Elementary Data Organization, Built in Data Types in C. Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic notations: Big Oh, Big Theta and Big Omega, Time-Space trade-off. Abstract Data Types (ADT) Arrays: Definition, Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order, Derivation of Index Formulae for 1-D, 2-D, 3-D and n-D Array Application of arrays, Sparse Matrices and their representations. Linked lists: Array Implementation and Pointer Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List. Insertion, Deletion, Traversal, Polynomial Representation and Addition Subtraction & Multiplications of Single variable & two variables Polynomial.

UNIT 2: Stacks and Queues: Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression, Iteration and Recursion-Principles of recursion, Tail recursion, Removal of recursion Problem solving using iteration and recursion with examples such as binary search, Fibonacci numbers, and Hanoi towers. Tradeoffs between iteration and recursion. Queues: Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues in C, Dequeue and Priority Queue.

UNIT 3: Searching & Sorting: Concept of Searching, Sequential search, Index Sequential Search, Binary Search. Concept of Hashing & Collision resolution Techniques used in Hashing. Sorting: Insertion Sort, Selection, Bubble Sort, Quick Sort, Merge Sort, Heap Sort and Radix Sort.

UNIT 4: Trees & Graphs: Basic terminology used with Tree, Binary Trees, Binary Tree Representation: Array Representation and Pointer (Linked List) Representation, Binary Search Tree, Strictly Binary Tree, Complete Binary Tree. Extended Binary Trees, Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Operation of Insertion, Deletion, Searching & Modification of data in Binary Search. Threaded Binary trees, Traversing Threaded Binary trees. Huffman coding using Binary Tree. Concept & Basic Operations for AVL Tree, B Tree & Binary Heaps. Graphs: Terminology used with Graph, Data Structure for Graph Representations: Adjacency Matrices, Adjacency List, Adjacency. Graph Traversal: Depth

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First Search and Breadth First Search, Connected Component, Spanning Trees, Minimum Cost Spanning Trees: Prim's and Kruskal algorithm. Transitive Closure and Shortest Path algorithm: Warshall Algorithm and Dijkstra Algorithm.

Course Outcomes (COs): After completing this course, students will be able to:

CO1: Understand the fundamental concepts of data structures and analyse algorithms using time and space complexity.

CO2: Apply linear data structures such as arrays, linked lists, stacks, and queues along with their operations.

CO3: Analyse and implement non-linear data structures such as trees and graphs with appropriate operations.

CO4: Apply suitable searching and sorting algorithms to solve computational problems efficiently.

CO5: Understand and apply different problem-solving paradigms to design efficient algorithms.

Textbooks:

1. Fundamentals of data structures – Ellis Horowitz and Sartaj Sahni
2. Aaron M. Tenenbaum, Yedidyah Langsam and Moshe J. Augenstein, "Data Structures Using C and C++", PHI Learning Private Limited, Delhi India.
3. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publications Pvt Ltd Delhi India.
4. Lipschutz, "Data Structures" Schaum's Outline Series, Tata McGraw-hill Education (India) Pvt. Ltd.
5. Thareja, "Data Structure Using C" Oxford Higher Education.
6. AK Sharma, "Data Structure Using C", Pearson Education India.
7. Rajesh K. Shukla, "Data Structure Using C and C++" Wiley Dreamtech Publication.
8. Michael T. Goodrich, Roberto Tamassia, David M. Mount "Data Structures and Algorithms in C++", Wiley India.

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Data Structure Lab

Course Code: AID251

Credit: 0+1

Experiments

Write C Programs to illustrate the concept of the following:

1. Sorting Algorithms-Non-Recursive.
2. Sorting Algorithms-Recursive.
3. Searching Algorithms.
4. Implementation of Stack using Array.
5. Implementation of Queue using Array.
6. Implementation of Circular Queue using Array.
7. Implementation of Stack using Linked List.
8. Implementation of Queue using Linked List.
9. Implementation of Circular Queue using Linked List.
10. Implementation of Tree Structures, Binary Tree, Tree Traversal, Binary Search Tree, Insertion and Deletion in BST.
11. Graph Implementation, BFS, DFS, Minimum cost spanning tree, shortest path algorithm

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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Database Management Systems

Course Code: AID202

Credit: 3+0

Course Objective: This course provides a fundamental understanding of Database Management Systems, covering both theoretical and practical aspects. It enables students to design and implement databases, apply core concepts effectively, and understand database systems using functional and procedural approaches.

Syllabus

UNIT 1: Introduction: An Overview of Database Management System, Database System vs File System, Database System Concept and Architecture, Data Model Schema and Instances, Data Independence and Database Language and Interfaces, Data Definitions Language, DML, Overall Database Structure.

Data Modelling using Entity Relationship Model: ER Model Concepts, Notation for ER Diagram, Mapping Constraints, Keys, Concepts of Super Key, Candidate Key, Primary Key, Generalization, Aggregation, Reduction of An ER Diagrams to Tables, Extended ER Model, Relationship of Higher Degree.

UNIT 2: Relational Data Model and Language: Relational Data Model Concepts, Integrity Constraints, Entity Integrity, Referential Integrity, Keys Constraints, Domain Constraints, Relational Algebra, Relational Calculus, Tuple and Domain Calculus.

Introduction on SQL: Characteristics of SQL, Advantage of SQL. SQL Data Type and Literals. Types of SQL Commands. SQL Operators and their Procedure. Tables, Views and Indexes. Queries and Sub Queries. Aggregate Functions. Insert, Update and Delete Operations, Joins, Unions, Intersection, Minus, Cursors, Triggers, Procedures in SQL/PL SQL.

UNIT 3: Database Design & Normalization: Functional Dependencies, Normal Forms, First, Second, Third Normal Forms, BCNF, Inclusion Dependence, Loss Less Join Decompositions, Normalization using FD, MVD, and JDS, Alternative Approaches to Database Design.

UNIT 4: Transaction Processing Concept: Transaction System, Testing of Serializability, Serializability of Schedules, Conflict & View Serializable Schedule, Recoverability, Recovery from Transaction Failures, Log Based Recovery, Checkpoints, Deadlock Handling.

Distributed Database: Distributed Data Storage, Concurrency Control, Directory System. Concurrency Control Techniques: Concurrency Control, Locking Techniques for Concurrency Control, Time Stamping Protocols for Concurrency Control, Validation Based Protocol, Multiple Granularity, Multi Version Schemes, Recovery with Concurrent Transaction, Case Study of Oracle.

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Course Outcomes (COs): The students are expected to be able to demonstrate the following knowledge, skills, and attitudes after completing this course.

C01: Define the basic concepts of database management systems.

C02: Discuss various database design techniques for developing database systems.

C03: Discuss various techniques for having control over concurrent database transactions.

C04: Discuss various advanced topics on database management systems

C05: Understand various transaction processing, concurrency control mechanisms and database protection mechanisms.

C06: Demonstrate an understanding of normalization theory and apply such knowledge to the normalization of a database.

Textbooks:

1. Date C J, An Introduction to Database Systems, Addison Wesley
2. Korth, Silbertz, Sudarshan, Database Concepts, McGraw Hill
3. Elmasri, Navathe, Fundamentals of Database Systems, Addison Wesley
4. O'Neil, Databases, Elsevier Pub.
5. Leon & Leon, Database Management Systems, Vikas Publishing House
6. Bipin C. Desai, An Introduction to Database Systems, Galgotia Publications
7. Majumdar & Bhattacharya, Database Management System, TMH
8. Ramkrishnan, Gehrke, Database Management System, McGraw Hill
9. Kroenke, Database Processing Fundamentals, Design and Implementation, Pearson Education.

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Database Management System Lab

Course Code: AID252

Credit: 0+1

Experiments

1. Write SQL queries using SQL operators.
2. Write SQL query using character, number, date and group functions.
3. Write SQL queries for relational algebra.
4. Write SQL queries for extracting data from more than one table.
5. Write SQL queries for sub queries, nested queries.
6. Write a program using PL/SQL.
7. Concepts for ROLL BACK, COMMIT & CHECK POINTS 8. Create VIEWS, CURSORS and TRGGERS & write ASSERTIONS.
8. Create FORMS and REPORTS.
9. Design of tables by normalization and dependency analysis 11. Writing application software with a host language interface.
10. J. D. Ulman, Principles of Database and Knowledge base System, Computer Science Press. 11. Maheshwari Jain. DBMS: Complete Practical Approach, Firewall Media, New Delhi

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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Software Engineering

Course Code: AID203

Credit: 3+0

Course Objectives: This course provides understanding of software development processes, design, testing, and maintenance, covering development models, requirement analysis, project management, and quality assurance to build reliable and maintainable software systems.

Syllabus

UNIT 1: Software Process– Introduction, S/W Engineering Paradigm, Life Cycle Models (Waterfall, Incremental, Spiral, Evolutionary, Prototyping), Software Requirements – Functional and Non-Functional–Software Document– Requirement Engineering Process–Feasibility Studies – Software Prototyping–Prototyping in Software, Process–Data–Functional and Behavioral Models– Structured Analysis And Data Dictionary.

UNIT 2: Basic Concept of Software Design, Architectural Design, Low Level Design: Modularization, Design Structure Charts, Pseudo Codes, Flow Charts, Coupling and Cohesion Measures, Design Strategies: Function Oriented Design, Object Oriented Design, Top-Down and Bottom-Up Design. Software Measurement and Metrics: Various Size Oriented Measures: Halstead's Software Science, Function Point (FP) Based Measures, Cyclomatic Complexity Measures: Control Flow Graph.

UNIT 3: Software Testing – Taxonomy of S/W Testing Levels - Black Box Testing – Testing Boundary Conditions – Structural Testing -- Regression Testing– S/W Testing Strategies, Unit Testing, Integration Testing, Validation Testing, System Testing and Debugging.

UNIT 4: Measures and Measurements – Zipf's Law, Software Cost Estimation – Function Point Models, COCOMO Model. Delphi Method – Scheduling – Earned Value Analysis – Error Tracking – Software Configuration Management – Program Evolution Dynamics – Software Maintenance – Project Planning – Project Scheduling– Risk Management – Case Tools

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Analyse software life cycle models and requirement engineering processes to define requirements and assess feasibility.

CO2: Design software systems using architectural and low-level design techniques with appropriate design strategies and modular principles.

CO3: Evaluate software using measurement metrics such as Halstead's metrics, Function Point analysis, and cyclomatic complexity.

CO4: Apply software testing techniques and strategies, including unit, integration, validation, and system testing.

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C05: Utilize software project management practices including cost estimation, scheduling, risk management, and maintenance.

Textbooks:

1. R. S. Pressman, "Software Engineering - A practitioner's approach", 3rd Edition, McGraw Hill International editions, 1992.
2. IAN Sommerville, Software Engineering, Pearson Education Asia, VI Edition, 2000.
3. Pankaj Jalote, "An Integrated Approach to Software Engineering", Springer Verlag, 1997.

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Discrete Mathematics

Course Code: AID204

Credit: 3+0

Course Objectives: This course builds a foundation in discrete mathematics, develops logical and problem-solving skills, and covers combinatorics, graph theory, and algebraic structures for algorithm design and competitive exams.

Syllabus

UNIT 1: Mathematical Logic: Propositional Logic: Statements, Logical Connectives, Truth Tables, Tautology, Contradiction, Logical Equivalence and Implication, Normal Forms: CNF, DNF, Predicate Logic: Quantifiers, Nested Quantifiers, Inference Theory and Proof Techniques, Mathematical Induction.

UNIT 2: Set Theory and Relations: Sets, Subsets, Power Set, Venn Diagrams, Operations on Sets, Relations: Properties (Reflexive, Symmetric, Transitive), Equivalence Relations and Classes, Partial Order Relations and Hasse Diagrams, Functions: Types (Injective, Surjective, Bijective).

UNIT 3: Combinatory and Counting: Basic Counting Principles, Permutations and Combinations, Binomial Theorem, Pigeonhole Principle, Inclusion-Exclusion Principle, Recurrence Relations and their solutions

Unit 4: Graph Theory: Basics of Graphs: Types, Terminology, Degree of Vertex, Paths, Cycles, Connected Components, Trees: Properties, Spanning Trees, Graph Traversal (BFS, DFS), Planar Graphs, Graph coloring, Shortest Path Algorithms (Basics).

Algebraic Structures: Groups: Definition, Properties, Examples, Subgroups and Cyclic Groups, Rings and Fields (Basic Concepts).

Course Outcomes (COs): After completing this course, students will be able to:

CO1: Understand and apply principles of mathematical logic and proof techniques.

CO2: Analyze sets, relations, and functions for solving computational problems.

CO3: Apply combinatorial techniques and recurrence relations in problem-solving.

CO4: Analyze graph structures and apply graph algorithms in various applications.

CO5: Understand algebraic structures and Boolean algebra for computing and digital systems.

Textbooks:

1. Discrete Mathematics and Its Applications – Kenneth H. Rosen
2. Elements of Discrete Mathematics – C. L. Liu
3. Discrete Mathematical Structures – Kolman, Busby & Ross

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Theory of Computation

Course Code: AID205

Credit: 3+0

Course Objective: Students should be able to understand the fundamental models of computation, including finite automata and Turing machines, apply these concepts in algorithm design, and analyse the theoretical limits of computation.

Syllabus

UNIT 1: Alphabets, Strings and Languages, Automata and Grammars, Deterministic Finite Automata (DFA) Formal Definition, Simplified Notation: State Transition Graph, Transition Table, Language of DFA, Nondeterministic Finite Automata (NFA), NFA with Epsilon Transition, Equivalence of NFA and DFA, Minimization of Finite Automata, Myhill-Nerode Theorem.

UNIT 2: Regular Expression (RE), Definition, Operators of Regular Expression and their Precedence, Algebraic Laws for Regular Expressions, Kleen's Theorem, Regular Expression to FA, DFA to Regular Expression, Arden Theorem, Non-Regular Languages, Pumping Lemma for Regular Languages. Application of Pumping Lemma, Closure Properties of Regular Languages, Decision Properties of Regular Languages, FA with Output: Moore and Mealy Machine, Equivalence of Moore and Mealy Machine, Applications and Limitation of FA.

UNIT 3: Context Free Grammar (CFG) and Context Free Languages (CFL), Definition, Examples, Derivation, Derivation Trees, Ambiguity in Grammar, Inherent Ambiguity, Ambiguous to Unambiguous CFG, Useless Symbols, Simplification of CFGs, Normal Forms for CFGs: CNF and GNF, Closure Properties of CFLs, Decision Properties of CFLs: Emptiness, Finiteness and Membership, Pumping Lemma for CFLs. Push Down Automata (PDA): Description and Definition, Instantaneous Description, Language of PDA, Acceptance by Final State, Acceptance by Empty Stack, Deterministic PDA, Equivalence of PDA and CFG, CFG to PDA and PDA to CFG, Two Stack PDA.

UNIT 4: Turing Machines (TM), Basic Model, Definition and Representation, Instantaneous Description, Language Acceptance by TM, Variants of Turing Machine, TM as Computer of Integer Functions, Universal TM, Church's Thesis, Recursive and Recursively Enumerable Languages, Halting Problem, Introduction to Undecidability, Undecidable Problems about TMs. Post Correspondence Problem (PCP), Modified PCP, Introduction to Recursive Function Theory.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Analyse and design finite automata, pushdown automata, Turing machines, formal languages, and grammars.

CO2: Demonstrate the understanding of key notions, such as algorithm, computability, decidability, and complexity through problem solving.

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C03: Prove the basic results of the Theory of Computation. State and explain the relevance of the Church Turing thesis.

C04: To solve various problems of applying normal form techniques, push down automata and Turing Machines.

C05: To Construct context free grammar for various languages.

C06: To Design Finite Automata for different Regular Expressions and Languages.

Textbooks:

1. Micheal Sipser, "Introduction to the Theory of Computation", Thomson Learning
2. Hopcroft, Ullman, "Introduction to Automata Theory, Languages and Computation", Pearson Education.
3. Peter Linz, "An Introduction to Formal Language and Automata", Narosa Publishing house
4. H R. Lewis and Christos H. Papadimitriou, "Elements of the theory of Computation", PHI Ltd

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Operating Systems

Course Code: AID206

Credit: 3+0

Course Objective: Students will learn about processes and processor management, concurrency and synchronization, memory management schemes, and secondary storage management, security and protection.

Syllabus

UNIT 1: Introduction: Operating system and functions, Classification of Operating systems- Batch, Interactive, Time-sharing, Real-Time System, Multiprocessor Systems, Multiuser Systems, Multi-process Systems, Multithreaded Systems, Operating System Structure- Layered structure, System Components, Operating System services, Reentrant Kernels, Monolithic and Microkernel Systems.

UNIT 2: Concurrent Processes: Process Concept, Principle of Concurrency, Producer / Consumer Problem, Mutual Exclusion, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Test and Set operation; Classical Problem in Concurrency- Dining Philosopher Problem, Sleeping Barber Problem; Inter Process Communication models and Schemes, Process generation.

UNIT 3: CPU Scheduling & Disk Scheduling: Scheduling Concepts, Performance Criteria, Process States, Process Transition Diagram, Schedulers, Process Control Block (PCB), Process address space, Process identification information, Thread and their management, Scheduling Algorithms, Multiprocessor Scheduling. Deadlock: System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock. Disk Scheduling: I/O devices, and I/O subsystems, I/O buffering, Disk storage and disk scheduling.

UNIT 4: Memory Management: Basic bare machine, Resident monitor, Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Protection schemes, Paging, Segmentation, Paged segmentation, Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing, Cache memory organization, Locality of reference.

Course Outcome (COs): At the end of course, the student will be able to:

CO1: Understand the structure and functions of the OS.

CO2: Learn about Processes, Thread and Scheduling algorithms.

CO3: Understand the principles of concurrency and Deadlocks.

CO4: Learn various memory management schemes.

CO5: Study I/O management and File systems.

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Textbooks:

1. Silberschatz, Galvin and Gagne, "Operating Systems Concepts", Wiley.
2. Sibsankar Halder and Alex A Aravind, "Operating Systems", Pearson Education.
3. Harvey M Dietel, "An Introduction to Operating System", Pearson Education.
4. D M Dhamdhere, "Operating Systems: A Concept based Approach", 2nd Edition, TMH.
5. William Stallings, "Operating Systems: Internals and Design Principles", 6th Edition, Pearson Education.

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Operating System Lab

Course Code: AID 254

Credit: 0+1

1. Study of hardware and software requirements of different operating systems (UNIX, LINUX, WINDOWS XP/ WINDOWS 7/8/10)
2. Execute various UNIX system calls for
 - a. Process management
 - b. File management
 - c. Input/output Systems calls
3. Implement CPU Scheduling Policies:
 - a. SJF
 - b. Priority
 - c. FCFS
 - d. Multi-level Queue
4. Implement file storage allocation technique:
 - a. Contiguous (using array)
 - b. Linked –list (using linked-list)
 - c. Indirect allocation (indexing)
5. Implementation of contiguous allocation techniques:
 - a. Worst-Fit
 - b. Best- Fit
 - c. First- Fit
6. Calculation of external and internal fragmentation
 - a. Free space list of blocks from system
 - b. List process file from the system
7. Implementation of compaction for the continually changing memory layout and calculate total movement of data.
8. Implementation of resource allocation graph (RAG).
9. Implementation of Banker's algorithm.
10. Conversion of resource allocation graph (RAG) to wait for graph (WFG) for each type of method used for storing graphs.
11. Implement the solution for the Bounded Buffer (producer-consumer) problem using inter process communication techniques-Semaphores.
12. Implement the solutions for Readers-Writer's problem using inter process communication technique -Semaphore.

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

Syllabus of B.Tech. Computer Science & Engineering-Artificial Intelligence & Data Science

Data Science & Visualization Technique

Course Code: AID 207

Credit: 3+0

Course Objective: Students should be able to understand the fundamental concepts of data science and AI, use advanced visualization tools effectively, and create interactive and dynamic visualizations for data-driven insights.

Syllabus

UNIT 1: Introduction to Data Science and AI: Overview of Data Science and AI, Data Types and Data Sources, Introduction to Machine Learning, Data Preprocessing Techniques, Exploratory Data Analysis (EDA), Introduction to Python for Data Science, Tools and Libraries for Data Science, Case Studies in Data Science and AI, Ethics and Bias in Data Science, Applications of Data Science and AI.

UNIT 2: Data Analysis Techniques: Statistical Analysis for Data Science, Introduction to Regression Analysis, Introduction to Classification, Introduction to Clustering, Dimensionality Reduction Techniques, Time Series Analysis, Text Analytics Basics, Web Scraping for Data Collection.

UNIT 3: Data Visualization Techniques: Introduction to Data Visualization, Principles of Effective Visualization, Data Types and Representation, Visualization Tools, Exploratory Data Analysis (EDA), Data Cleaning and Preparation, Static vs. Interactive Visualizations. Interactive Visualization with D3.js, Geospatial Data Visualization, Network Visualization, Hierarchical and Tree Visualization, Time-Series Visualization, 3D and Virtual Reality Visualization, Customizing Visualizations with CSS and JavaScript, Advanced Chart Types and Graphs, Animation and Transition Effects, Dashboard Design Principles.

UNIT 4: Practical Applications: Predictive Analytics, Image Recognition Basics, Speech Recognition Basics, Natural Language Understanding Basics, Recommendation Systems Basics, Fraud Detection Basics, Basic Time Series Forecasting, Introduction to Autonomous Systems, Healthcare Analytics Fundamentals, Financial Analytics Basics. Visualizing Complex Datasets, Case Studies in Data Visualization, Project-based Learning, Collaborative Visualization Projects, Presentation and Critique of Visualization Projects, Final Visualization Project Showcase.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Understand the basic principles and concepts of data science.

CO2: Understand the concepts of advanced data science techniques.

CO3: Proficiency in using advanced visualization tools.

CO4: Ability to create interactive and dynamic visualizations.

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Textbooks:

1. "Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking" by Foster Provost and Tom Fawcett
2. "Data Science from Scratch: First Principles with Python" by Joel Grus
3. "Interactive Data Visualization for the Web" by Scott Murray
4. "D3.js in Action" by Elijah Meeks
5. "The Visual Display of Quantitative Information" by Edward R. Tufte
7. "Information Dashboard Design: Displaying Data for At-a-Glance Monitoring" by Stephen Few

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Data Science & Visualization Technique Lab

Course Code: AID 255

Credit: 0+1

Experiments:

1. Introduction to various Data Visualization tools
2. Basic Visualization in Python
3. Basic Visualization understanding Numpy and Pandas
4. Understanding process of EDA
5. Connecting to Data and preparing data for visualization in Tableau
6. Data Aggregation and Statistical functions in Tableau
7. Data Visualizations in Tableau
8. Basic Dashboards in Tableau

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

Syllabus of B.Tech. Computer Science & Engineering-Artificial Intelligence & Data Science

Computer Networks

Course Code: AID208

Credit: 3+0

Course Objective: This course covers layered communication models, data transmission across physical links, network and protocol design, and the analysis of key protocols and challenges in distributed networked systems.

Syllabus

UNIT 1: Introductory Concepts: Goals and applications of networks, Categories of networks, Organization of the Internet, ISP, Network structure and architecture (layering principles, services, protocols and standards), The OSI reference model, TCP/IP protocol suite, Network devices and components. Physical Layer: Network topology design, Types of connections, Transmission media, Signal transmission and encoding, Network performance and transmission impairments, switching techniques and multiplexing.

UNIT 2: Link Layer: Framing, Error Detection and Correction, Flow control (Elementary Data Link Protocols, Sliding Window protocols). Medium Access Control and Local Area Networks: Channel allocation, multiple access protocols, LAN standards, Link layer switches & bridges (learning bridge and spanning tree algorithms).

UNIT 3: Network Layer: Point-to-point networks, Logical addressing, Basic internetworking (IP, CIDR, ARP, RARP, DHCP, and ICMP), Routing, forwarding and delivery, Static and dynamic routing, Routing algorithms and protocols, Congestion control algorithms, IPv6.

UNIT 4: Transport Layer: Process-to-process delivery, Transport layer protocols (UDP and TCP), Multiplexing, Connection management, Flow control and retransmission, Window management, TCP Congestion control, Quality of service. **Application Layer:** Domain Name System, World Wide Web and Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Remote login, Network management, Data compression, Cryptography – basic concepts.

Course Outcome (COs): At the end of course, the student will be able to

CO1: Explain basic concepts, OSI reference model, services and role of each layer of OSI model and TCP/IP, networks devices and transmission media, Analog and digital data transmission.

CO2: Apply channel allocation, framing, error and flow control techniques.

CO3: Describe the functions of Network Layer i.e., Logical addressing, subnetting & Routing Mechanism.

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C04: Explain the different Transport Layer functions i.e., Port addressing, Connection Management, Error control and Flow control mechanism.

C05: Explain the functions offered by session and presentation layer and their Implementation.

C06: Explain the different protocols used at application layer i.e., HTTP, SNMP, SMTP, FTP, TELNET and VPN.

Textbooks:

1. Behrouz Forouzan, "Data Communication and Networking", McGraw Hill
2. Andrew Tanenbaum "Computer Networks", Prentice Hall.
3. William Stallings, "Data and Computer Communication", Pearson.
4. Kurose and Ross, "Computer Networking- A Top-Down Approach", Pearson.
5. Peterson and Davie, "Computer Networks: A Systems Approach", Morgan Kaufmann
6. W. A. Shay, "Understanding Communications and Networks", Cengage Learning.
7. D. Comer, "Computer Networks and Internets", Pearson.
8. Behrouz Forouzan, "TCP/IP Protocol Suite", McGraw Hill.

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Computer Network Lab

Course Code: AID256

Credit: 0+1

Experiments

1. To create scenarios and study the performance of CSMA/CD protocol through simulation.
2. To create scenarios and study the performance of token bus and token ring protocols through simulation.
3. Implementation of Error detection and correction algorithms.
4. Implementation and study of 1-bit sliding window viz., stop and wait protocol.
5. Implementation and study of Go back-N protocol.
6. Implementation and study of selective repeat protocol.
7. To get the MAC or Physical address of the system using Address Resolution Protocol.
8. Implementation of distance vector routing algorithm.
9. Implementation of link state routing algorithm.
10. To write a client-server application for chat using TCP.
11. To write a C program to develop a DNS client server to resolve the given hostname.

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

Syllabus of B.Tech. Computer Science & Engineering-Artificial Intelligence & Data Science

Probability and Statistics for Data Science

Course Code: AID209

Credit: 3+0

Course objectives: The purpose of this course is to introduce the fundamental rules of Probability, discrete and continuous distributions, and statistical methods that are most commonly used in Computer Science and Engineering. Students will be introduced to stochastic processes, multivariate analysis and statistical inference methods and will apply the theory and methods to the evaluation of queuing systems and computation of their vital characteristics.

Syllabus

UNIT 1: Descriptive Statistics: Introduction to statistics, collection, and tabulation of data, different types of charts and graphs. Measures of central tendency, Frequency distributions, Histograms, Box plots, Scatter plots, Quartiles and percentiles, Arithmetic mean, Geometric mean, harmonic mean, median and mode. Measures of dispersion: Range, mean deviation, Standard deviation, co-efficient of variation, moments, Skewness and Kurtosis, Covariance, correlation and regression coefficient, line of regression.

UNIT 2: Probability and Distribution: Random experiment, Probability rules, Conditional probability, Bayes' theorem, Single Random Variables: Discrete and Continuous, Probability distribution function, Probability mass and density functions, mathematical expectation and variance. Covariance, MGF, Multiple Random variables: Discrete and Continuous, Joint probability distribution, Marginal probability density functions, conditional probability distribution function and density functions. Probability distributions (Bernoulli, Binomial, Poisson, Normal distribution), Sampling distribution (Chi-square, t and F-distribution), Central Limit Theorem.

UNIT 3: Estimation and Testing of Hypothesis: Sampling distributions, Estimation, point estimation, Interval estimation, Maximum likelihood estimation, Method of moments, Bayesian estimation, Confidence intervals, Prediction intervals, Estimating sample statistics, Hypothesis testing (Z-test, t-test), A/B testing, Chi-square test, ANOVA,

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Non-parametric tests, Type I & Type II error, Level of Significance and Power and sample size calculations.

UNIT 4: Regression Analysis and Multivariate Techniques: Notation; Ordinary least squares, Regression to the mean, Linear regression, Multiple regression, Generalized linear models, Random vector, Bivariate Normal distribution, Quadratic form, Wishart distribution, Hotelling T^2 -statistics, Multivariate analysis technique (MANOVA), Multivariate Regression, Principal Component analysis (PCA), Canonical correlation.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Represents the rudiments of Data Science.

CO2: Understand the concept of Probability and one-dimensional random variable.

CO3: Demonstrate the rules of probability and statistics for understanding the nature of data.

CO4: Understand the importance of discrete and continuous distributions.

CO5: Perform test of hypothesis as well as calculate confidence interval for the population parameter.

CO6: Understand the importance of multivariate data analysis.

Textbooks:

1. Milton. J. S. and Arnold. J.C., "Introduction to Probability and Statistics", Tata McGraw Hill, 4th Edition, 2007.
2. Johnson. R.A. and Gupta. C.B., "Miller and Freund's Probability and Statistics for Engineers", Pearson Education, Asia, 7th Edition, 2007.
3. Statistics for Management, Seventh Edition, by Richard I. Levin, David S. Rubin, Pearson Chandrasekaran, G. Kavitha, "Probability, Statistics, Random Processes and Queuing Theory", Dhanam Publications, 2014.
4. Raj Kumar Bansal, Ashok Kumar Goel, Manoj Kumar Sharma, "MATLAB and its Applications in Engineering", Pearson Publication, Second Edition, 2016.
5. An Introduction to Categorical Data Analysis. Agresti, A. (2012). John Wiley & sons

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6. Spiegel. M.R., Schiller. J. and Srinivasan. R.A., "Schaum's Outline of Theory and Problems of Probability and Statistics", Tata McGraw Hill Edition, 2004.

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Machine Learning

Course Code: AID301

Credit: 3+0

Course Objective: Students should be able to apply fundamental machine learning concepts, perform effective data pre-processing, and interpret and communicate model results in real-world contexts.

Syllabus

UNIT 1: Introduction to Machine Learning: Introduction to machine learning, Types of machine learning algorithms, Supervised learning, Unsupervised learning, Semi-supervised learning, Reinforcement learning, Bias-variance tradeoff, Overfitting and Underfitting, Model evaluation metrics, Cross-validation techniques, Ethics in machine learning, Future trends in machine learning

UNIT 2: Data Preprocessing and Feature Engineering: Data preprocessing techniques, Data cleaning, Missing value imputation, Outlier detection and treatment, Data transformation, Feature scaling, Feature selection, Dimensionality reduction, Encoding categorical variables, Handling imbalanced datasets, Time series data preprocessing, Feature engineering strategies

UNIT 3: Supervised Learning Algorithms: Linear regression, Polynomial regression, Ridge regression, Lasso regression, Logistic regression, Decision trees, Random forests, Gradient boosting, Support vector machines, k-Nearest neighbors, Naive Bayes, Neural networks.

Unsupervised Learning Algorithms: K-means clustering, Hierarchical clustering, DBSCAN, Gaussian mixture models, Dimensionality reduction techniques (PCA, t-SNE), Independent component analysis (ICA), Non-negative matrix factorization (NMF), Autoencoders, Association rule learning, Anomaly detection methods, Density-based clustering algorithms, Spectral clustering

UNIT 4: Advanced Topics in Machine Learning: Introduction to deep learning, Convolutional neural networks (CNNs), Recurrent neural networks (RNNs), Long Short-Term Memory (LSTM) networks, Generative adversarial networks (GANs), Reinforcement learning algorithms, Policy gradients, Q-learning, Actor-critic methods, Transfer learning, Model evaluation and hyperparameter tuning, Bias and fairness in machine learning

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Explain theory underlying machine learning

CO2: Construct algorithms to learn linear and non-linear models

CO3: Implement data clustering algorithms

CO4: Construct algorithms to learn tree and rule-based models

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C05: Apply reinforcement learning techniques

Textbooks:

1. "Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller and Sarah Guido.
2. "Pattern Recognition and Machine Learning" by Christopher M. Bishop
3. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron
4. "Machine Learning: A Probabilistic Perspective" by Kevin P. Murphy
5. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville

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Machine Learning Lab

Course Code: AID351

Credit: 0+1

Experiments

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.
3. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
4. Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
5. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.
6. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Built-in python classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.
7. Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using the standard Heart Disease Data Set. You can use Python ML library classes/API.
8. Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using the k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Python ML library classes/API in the program.
9. Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem.
10. Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select the appropriate data set for your experiment and draw graphs.
11. Carry out the performance analysis of classification algorithms on a specific dataset.

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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Design and Analysis of Algorithms

Course Code: AID302

Credit: 3+0

Course Objective: The goal of this course is to provide a solid background in the design and analysis of the major classes of algorithms. At the end of the course students will be able to develop their own versions for a given computational task and to compare and contrast their performance.

Syllabus

UNIT 1: Introduction: Algorithms, Analyzing Algorithms, Complexity of Algorithms, Growth of Functions, Performance Measurements, Sorting and Order Statistics - Shell Sort, Quick Sort, Merge Sort, Heap Sort, Comparison of Sorting Algorithms, Sorting in Linear Time.

UNIT 2: Advanced Data Structures: Red-Black Trees, B – Trees, Binomial Heaps, Fibonacci Heaps, Tries, Skip List.

UNIT 3: Divide and Conquer with Examples: Such as Sorting, Matrix Multiplication, Convex Hull and Searching. Greedy Methods, Optimal Reliability Allocation, Knapsack, Minimum Spanning Trees – Prim’s and Kruskal’s Algorithms, Single Source Shortest Paths - Dijkstra’s and Bellman Ford Algorithms.

UNIT 4: Dynamic Programming with Examples Such as Knapsack. All Pair Shortest Paths – Warshal’s and Floyd’s Algorithms, Resource Allocation Problem. Backtracking, Branch and Bound with Examples Such as Travelling Salesman Problem, Graph Coloring, n-Queen Problem, Hamiltonian Cycles and Sum of Subsets.

Course Outcomes (COs): At the end of course, the student will be able to:

CO1: Design new algorithms, prove them correct, and analyse their asymptotic and absolute runtime and memory demands.

CO2: Find an algorithm to solve the problem (create) and prove that the algorithm solves the problem correctly (validate).

CO3: Understand the mathematical criterion for deciding whether an algorithm is efficient, and know many practically important problems that do not admit any efficient algorithms.

CO4: Apply classical sorting, searching, optimization and graph algorithms.

CO5: Understand basic techniques for designing algorithms, including the techniques of recursion, divide- and-conquer, and greedy.

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Textbooks:

1. Thomas H. Cormen, Charles E. Leiserson and Ronald L. Rivest, "Introduction to Algorithms", Prentice Hall of India.
2. E. Horowitz & S Sahni, "Fundamentals of Computer Algorithms",
3. Aho, Hopcraft, Ullman, "The Design and Analysis of Computer Algorithms" Pearson Education, 2008.
4. LEE "Design & Analysis of Algorithms (POD)", McGraw Hill
5. Richard E. Neapolitan "Foundations of Algorithms" Jones & Bartlett Learning
6. Jon Kleinberg and Éva Tardos, Algorithm Design, Pearson, 2005.
7. Michael T Goodrich and Roberto Tamassia, Algorithm Design: Foundations, Analysis, and Internet Examples, Second Edition, Wiley, 2006.
8. Harry R. Lewis and Larry Denenberg, Data Structures and Their Algorithms, Harper Collins, 1997
9. Robert Sedgewick and Kevin Wayne, Algorithms, fourth edition, Addison Wesley, 2011.
10. Harsh Bhasin," Algorithm Design and Analysis", First Edition, Oxford University Press.
11. Gilles Brassard and Paul Bratley, Algorithmics: Theory and Practice, Prentice Hall, 1995.

Design and Analysis of Algorithm Lab

Course Code: AID352

Credit: 0+1

Experiments

1. Program for Recursive Binary & Linear Search.
2. Program for Heap Sort.
3. Program for Merge Sort.
4. Program for Selection Sort.
5. Program for Insertion Sort.
6. Program for Quick Sort.
7. Knapsack Problem using Greedy Solution
8. Perform Travelling Salesman Problem
9. Find Minimum Spanning Tree using Kruskal's Algorithm
10. Implement N Queen Problem using Backtracking
11. Sort a given set of n integer elements using Quick Sort method and compute its time complexity. Run the program for varied values of $n > 5000$ and record the time taken to sort. Plot a graph of the time taken versus a non-graph sheet. The elements can be read from a file or can be generated using the random number generator. Demonstrate using Java how the divide and- conquer method works along with its time complexity analysis: worst case, average case and best case.
12. Sort a given set of n integer elements using the Merge Sort and compute its time complexity. Run the program for varied values of $n > 5000$, and record the time taken to sort. Plot a graph of the time taken versus a non-graph sheet. The elements can be read from a file or can be generated using the random number generator. Demonstrate how the divide and- conquer method works along with its time complexity analysis: worst case, average case and best case.
13. Implement the 0/1 Knapsack problem using
 - a. Dynamic Programming method
 - b. Greedy method
14. From a given vertex in a weighted connected graph, find shortest paths to other vertices using Dijkstra's algorithm.
15. Find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm. Use Union-Find algorithms in your program.
16. Find Minimum Cost Spanning Tree of a given undirected graph using Prim's algorithm.
17. Write programs to-
 - a. Implement All-Pairs Shortest Paths problem using Floyd's algorithm.
 - b. Implement Travelling Sales Person problem using Dynamic programming.
18. Design and implement to find a subset of a given set $S = \{S_1, S_2, \dots, S_n\}$ of n positive integers whose SUM is equal to a given positive integer d. For example, if $S = \{1, 2,$

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5, 6, 8} and $d = 9$, there are two solutions {1,2,6} and {1,8}. Display a suitable message, if the given problem instance doesn't have a solution.

19. Design and implement to find all Hamiltonian Cycles in a connected undirected Graph G of n vertices using backtracking principle.

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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Artificial Intelligence

Course Code: AID303

Credit: 3+0

Course Objective: This course sheds light on the fundamentals of Artificial Intelligence and its applications in various areas. The student will learn to apply knowledge representation techniques and problem-solving strategies to common AI applications.

Syllabus

UNIT 1: Introduction: Definition – Future of Artificial Intelligence – Characteristics of Intelligent Agents– Typical Intelligent Agents – Problem Solving Approach to Typical AI problems.

UNIT 2: Problem Solving Methods: Problem solving Methods – Search Strategies- Uninformed – Informed – Heuristics – Local Search Algorithms and Optimization Problems – Searching with Partial Observations – Constraint Satisfaction Problems – Constraint Propagation – Backtracking Search – Game Playing – Optimal Decisions in Games – Alpha – Beta Pruning – Stochastic Games.

UNIT 3: Knowledge Representation: First Order Predicate Logic – Prolog Programming – Unification – Forward Chaining-Backward Chaining – Resolution – Knowledge Representation – Ontological Engineering-Categories and Objects – Events – Mental Events and Mental Objects – Reasoning Systems for Categories – Reasoning with Default Information.

UNIT 4: Software Agents & Applications: Architecture for Intelligent Agents – Agent communication – Negotiation and Bargaining – Argumentation among Agents – Trust and Reputation in Multi- agent systems. **Applications:** AI applications – Language Models – Information Retrieval- Information Extraction – Natural Language Processing – Machine Translation – Speech Recognition – Robot – Hardware – Perception – Planning – Moving.

Course Outcomes (COs): At the end of course, the student will be able:

CO1: To understand the basics of the theory and practice of Artificial Intelligence as a discipline and about intelligent agents.

CO2: Understand search techniques and gaming theory.

CO3: The student will learn to apply knowledge representation techniques and problem-solving strategies to common AI applications.

CO4: Students should be aware of techniques used for classification and clustering.

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Textbooks:

1. S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach", Prentice Hall, Third Edition, 2009.
2. Bratko, "Prolog: Programming for Artificial Intelligence", Fourth edition, Addison-Wesley Educational Publishers Inc., 2011.
3. M. Tim Jones, —Artificial Intelligence: A Systems Approach, Jones and Bartlett Publishers, Inc. First Edition, 2008
4. Nils J. Nilsson, —The Quest for Artificial Intelligence, Cambridge University Press, 2009.
5. William F. Clocksin and Christopher S. Mellish, Programming in Prolog: Using the ISO Standard, Fifth Edition, Springer, 2003. Gerhard Weiss, —Multi Agent Systems, Second Edition, MIT Press, 2013.

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Data Mining and Data Warehousing

Course Code: AID 304

Credit: 3+0

Course Objective: Students should be able to differentiate between OLTP and OLAP systems, design multidimensional schemas for data warehousing, and understand as well as apply key data mining functionalities and algorithms.

Syllabus

UNIT 1: Data Warehousing and Business Analysis: Data warehousing Components – Building a Data warehouse –Data Warehouse Architecture – DBMS Schemas for Decision Support – Data Extraction, Cleanup, and Transformation Tools –Metadata – reporting – Query tools and Applications – Online Analytical Processing (OLAP) – OLAP and Multidimensional Data Analysis.

UNIT 2: Data Mining: Data Mining Functionalities – Data Preprocessing – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization and Concept Hierarchy Generation- Architecture of Typical Data Mining Systems- Classification of Data Mining Systems. Association Rule Mining: - Efficient and Scalable Frequent Item set Mining Methods – Mining Various Kinds of Association Rules – Association Mining to Correlation Analysis – Constraint-Based Association Mining.

UNIT 3: Classification and Prediction: Issues Regarding Classification and Prediction – Classification by Decision Tree Introduction – Bayesian Classification – Rule Based Classification – Classification by Back propagation – Support Vector Machines – Associative Classification – Lazy Learners – Other Classification Methods – Prediction – Accuracy and Error Measures – Evaluating the Accuracy of a Classifier or Predictor – Ensemble Methods – Model Section.

UNIT 4: Cluster Analysis, Mining Object, Spatial, Multimedia, Text and Web Data: Types of Data in Cluster Analysis – A Categorization of Major Clustering Methods – Partitioning Methods – Hierarchical methods – Density-Based Methods – Grid-Based Methods – Model-Based Clustering Methods – Clustering High-Dimensional Data – Constraint-Based Cluster Analysis – Outlier Analysis. Mining Object, Spatial, Multimedia, Text and Web Data: Multidimensional Analysis and Descriptive Mining of Complex Data Objects – Spatial Data Mining – Multimedia Data Mining – Text Mining – Mining the World Wide Web.

Course Outcomes (COs): At the end of course, the student will be able to:

CO1: Design and implement a data warehouse architecture, including schema modelling (star/snowflake), ETL processes, metadata management, and OLAP-based analytical reporting.

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C02: Apply data pre-processing techniques such as cleaning, integration, transformation, reduction, and discretization to prepare real-world datasets for effective data mining.

C03: Analyse and implement association rule mining algorithms to discover frequent patterns, correlations, and constraints in large transactional datasets.

C04: Develop and evaluate classification and prediction models using decision trees, Bayesian methods, SVMs, neural networks, and ensemble techniques, with appropriate accuracy and error metrics.

C05: Perform clustering and advanced data mining tasks on complex datasets (spatial, multimedia, text, and web), including outlier detection and high-dimensional data analysis.

Textbooks:

1. Jiawei Han, Micheline Kamber and Jian Pei "Data Mining Concepts and Techniques", Third Edition, Elsevier, 2011.
2. Alex Berson and Stephen J. Smith "Data Warehousing, Data Mining & OLAP", Tata McGraw – Hill Edition, Tenth Reprint 2007.
3. K.P. Soman, Shyam Diwakar and V. Ajay "Insight into Data mining Theory and Practice", Easter Economy Edition, Prentice Hall of India, 2006.
4. G. K. Gupta "Introduction to Data Mining with Case Studies", Easter Economy Edition, Prentice Hall of India, 2006.
5. Pang-Ning Tan, Michael Steinbach and Vipin Kumar "Introduction to Data Mining", Pearson.

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AI, Ethics and Society

Course Code: AID305

Credit: 3+0

Course Objective: This course introduces ethical principles and societal impacts of AI, examines bias, fairness, and privacy issues, and develops understanding of AI's role, policies, and regulations for responsible use.

Syllabus

UNIT 1: Foundations of AI Ethics

Introduction to AI and Ethics: Definition and scope of AI ethics, historical context of AI development, importance of ethical considerations in AI, ethical challenges in machine learning and automation, AI's impact on decision-making and human autonomy.

Ethical Theories and AI: Utilitarianism, deontology, virtue ethics, fairness, accountability, and transparency (FAT) in AI, principles of responsible AI development, ethical dilemmas in AI applications, trade-offs between privacy and utility.

UNIT 2: Bias, Fairness, and Privacy in AI

Bias and Fairness in AI: Sources of bias in AI models, algorithmic fairness and discrimination, mitigating bias in data and model training, fairness-aware machine learning techniques, real-world examples of biased AI decisions.

Privacy and Surveillance: Data privacy in AI systems, ethical concerns in data collection and processing, GDPR and AI regulations, AI-driven surveillance and its societal implications, privacy-preserving AI techniques (differential privacy, federated learning).

UNIT 3: Societal Impact and AI Governance

AI and Society: AI's impact on employment and the future of work, AI in healthcare, finance, and education, social and cultural implications of AI-driven automation, ethical concerns in AI-generated content and misinformation, AI and sustainability.

AI Governance and Regulation: Global AI policies and frameworks, role of government and industry in AI regulation, AI ethics guidelines by organizations (EU, IEEE, UNESCO), case studies of AI governance failures, ethical AI deployment in businesses and public services.

UNIT 4: Ethical AI Design and the Future of AI

Ethical AI Development and Human-Centered AI: Explainable AI (XAI) and interpretability, human-in-the-loop AI systems, designing AI for social good, ethical AI design principles, AI for accessibility and inclusion.

The Future of AI and Ethical Challenges: Emerging ethical challenges in AI (AGI, autonomous weapons, deepfake technology), ethical concerns in AI-driven decision-

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making, philosophical perspectives on AI consciousness, ensuring ethical AI development for future generations.

Course Outcomes (COs): Upon completion of the course, the students should be able to:

CO1: Introduce students to the ethical principles and societal impact of AI technologies.

CO2: Analyse bias, fairness, and privacy issues in AI models and data-driven decision-making.

CO3: Explore AI's role in society, including its impact on employment, governance, and cultural aspects.

CO4: Understand AI policies, regulations, and global ethical frameworks for responsible AI development.

CO5: Design AI systems that align with ethical considerations, transparency, and human values.

Textbooks:

1. Mark Coeckelbergh – *AI Ethics*, MIT Press, 2020.
2. Nick Bostrom – *Superintelligence: Paths, Dangers, Strategies*, Oxford University Press, 2014.
3. Virginia Eubanks – *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor*, St. Martin's Press, 2018.
4. Kate Crawford – *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*, Yale University

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Data Analytics using R Programming

Course Code: AID306

Credit: 3+0

Course Objective: Students should be able to perform data manipulation and analysis using statistical and machine learning techniques, and generate reports with graphical visualizations using R.

Syllabus

UNIT 1: Introduction to R & Programming Fundamentals: Installation, RStudio, and basic syntax. Data types: Vectors, Matrices, Factors, Lists, and Data Frames. Control structures: Loops (for, while, repeat), conditional statements (if, else). Writing user-defined functions.

UNIT 2: Data Manipulation & Data Visualization: Reading/writing data (CSV, TXT, Excel). Data manipulation with dplyr (select, filter, arrange, mutate, group_by). Handling missing data and data cleaning. apply family functions (apply, lapply, sapply, tapply). Descriptive statistics: Mean, median, mode, variance, skewness, Kurtosis. Data visualization: Basic plots and ggplot2 (histograms, box plots, scatter plots). Exploratory Data Analysis (EDA).

UNIT 3: Probability Distributions & Statistical Modelling: Probability distributions: Normal, Binomial, Poisson, Exponential, Weibull. Probability density functions (PDF) and cumulative distribution functions (CDF). Regression Model: Linear regression (lm function), Generalized Linear Model (glm function) and correlation. Hypothesis testing: t-tests, chi-square tests, ANOVA.

UNIT 4: Advanced Computing & Simulations in R: Generating random data/simulation, Monte Carlo integration/simulation, Optimization and root finding (Linear and non-linear optimization, quadratic optimization, Iteration Method, Bisection Method and Newton-Raphson Method), Numerical integration (Trapezoidal rule, Simpson's 1/3 and 3/8 rule, Waddle's rule), Solution of differential Equations (Taylor's Method, Euler's Method, Runge-Kutta Method).

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Gain basic knowledge on data science

CO2: Grasp data cleaning and reshaping in R.

CO3: Perform exploratory data analysis (EDA) and visualizing results.

CO4: Develop suitable models using machine learning techniques and to analyse its performance.

CO5: Identify the requirement and visualize the results.

CO6: Implementing simulation methods and statistical tests.

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Textbooks:

1. "An Introduction to Statistical Learning: with Applications in R" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.
2. "Applied Predictive Modelling" by Max Kuhn and Kjell Johnson
3. "Regression Modelling Strategies: With Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis" by Frank E. Harrell Jr.
4. Statistical Computing with R by Maria L. Rizzo.
5. Han, J., Kamber, M., Pei, J. Data mining concepts and techniques. Morgan Kaufmann. 2011
6. Carl Shan, Henry Wang, William Chen, Max Song. The Data Science Handbook: Advice and Insight from 25 Amazing Data Scientists. The Data Science Bookshelf. 2016.
7. Michael Akritas, " Probability & Statistics with R for Engineers and Scientists", 2nd Edition on, CRC Press, 2016.
8. James, G., Witten, D., T., Tibshirani, R. An Introduction to statistical learning with applications in R. Springer. 2013.

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Deep Learning & Neural Networks

Course Code: AID307

Credit: 3+0

Course Objective: This course provides a comprehensive understanding of deep learning and neural networks, develops practical skills for real-world applications, and enables students to evaluate, optimize, and fine-tune models effectively.

Syllabus

UNIT 1: Introduction to deep learning, Basics of neural networks, Perceptron, Activation functions, Gradient descent, Backpropagation, Overfitting and regularization, Hyperparameter tuning, Optimization algorithms, Loss functions, Model evaluation, Dropout regularization, Batch normalization, Learning rate schedules, Cross-validation, Early stopping

UNIT 2: Convolutional Neural Networks (CNNs) CNN architecture, Convolutional layers, Pooling layers, Transfer learning, Image classification, Object detection, Semantic segmentation, Style transfer, Data augmentation, Fine-tuning, Convolutional neural network visualization, Object localization, Adversarial training, Capsule networks, Autoencoders

UNIT 3: Recurrent Neural Networks (RNNs) RNN architecture, Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Sequence prediction, Natural Language Processing (NLP), Sentiment analysis, Named Entity Recognition (NER), Language translation, Text generation, Speech recognition, Attention mechanisms in RNNs, Bidirectional RNNs, Encoder-Decoder architectures, Reinforcement learning with RNNs, Reinforcement learning in dialogue systems

UNIT 4: Generative Adversarial Networks (GANs) GAN architecture, Generator and Discriminator networks, Training GANs, Applications in image generation and data synthesis, Conditional GANs, CycleGAN, Wasserstein GAN, Evaluation metrics, Ethical considerations, Variational Autoencoders (VAEs), GANs for anomaly detection, GANs for style transfer, GANs for super-resolution, Multi-agent GANs .Advanced Topics in Deep Learning Attention mechanisms, Transformers, Reinforcement learning, Deep reinforcement learning, Self-supervised learning, Meta-learning, Adversarial attacks and defenses, Explainable AI, Federated learning, AI fairness and bias, Quantum machine learning, Neuroevolution, Transfer learning across modalities, Capsule networks, Multi-task learning

Course Outcomes (COs): The students are expected to be able to demonstrate the following knowledge, skills and attitudes after completing this course.

CO1: Demonstrate proficiency in designing and training neural networks for real-world applications.

CO2: Apply deep learning techniques to solve complex problems in diverse domains

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C03: Analyze and interpret the performance of deep learning models.

Textbooks:

1. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, and Aaron Courville.
2. "Neural Networks and Deep Learning: A Textbook" by Charu C. Aggarwal.
3. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron.
4. "Deep Reinforcement Learning Hands-On" by Maxim Lapan.
5. "Python Deep Learning" by Ivan Vasilev and Daniel Slater

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Deep Learning & Neural Networks Lab

Course Code: AID353

Credit: 0+1

Experiments

1. Implement simple vector addition in TensorFlow.
2. Implement a regression model in Keras.
3. Implement a perceptron in TensorFlow/Keras Environment.
4. Implement a Feed-Forward Network in TensorFlow/Keras.
5. Implement an Image Classifier using CNN in TensorFlow/Keras.
6. Improve the Deep learning model by fine tuning hyper parameters.
7. Implement a Transfer Learning concept in Image Classification.
8. Using a pre trained model on Keras for Transfer Learning
9. Perform Sentiment Analysis using RNN
10. Implement an LSTM based Autoencoder in TensorFlow/Keras.
11. Image generation using GAN

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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Advanced Java Programming Lab

Course Code: AID354

Credit: 0+1

1. Write a JAVA program to display default value of all primitive data type of JAVA
2. Write a java program that displays the roots of a quadratic equation $ax^2+bx=0$. Calculate the discriminate D and basing on the value of D, describe the nature of the root.
3. Write a JAVA program to search for an element in a given list of elements using binary search mechanism.
4. Write a JAVA program to sort for an element in a given list of elements using bubble sort.
5. Write a JAVA program using String Buffer to delete, remove characters.
6. Write a JAVA program to implement class mechanism. Create a class, methods and invoke them inside the main method.
7. Write a JAVA program that implements method overloading.
8. Write a JAVA program to implement constructors.
9. Write a JAVA program to implement constructor overloading.
10. Write a JAVA program to implement Single Inheritance.
11. Write a JAVA program to implement multilevel Inheritance.
12. Write a JAVA program for abstract class to find areas of different shapes.
13. Write a JAVA program giving examples for “super” keywords.
14. Write a JAVA program to implement Interface. What kind of Inheritance can be achieved?
15. Write a JAVA program that implements Runtime polymorphism.
16. Write a JAVA program that describes exception handling mechanisms.
17. Write a JAVA program Illustrating Multiple catch clauses.
18. Write a JAVA program for creation of Java Built-in Exceptions.
19. Write a JAVA program for creation of User Defined Exception.
20. Write a JAVA program that imports and uses the user defined packages.
21. Without writing any code, build a GUI that displays text in label and image in an ImageView (use JavaFX).
22. Build a Tip Calculator app using several JavaFX components and learn how to respond to user interactions with the GUI.
23. Implement the programs using List Interface and its implemented classes.
24. Implement the programs using Set Interface and its implemented classes.
25. Implement the programs using Map Interface and its implemented classes.

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

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Computer Based Numerical and Statistical Technique

Course Code: AID308

Credit: 3+0

Course Objective: This course builds a foundation in numerical and statistical methods, develops computational skills for data analysis, and emphasizes accuracy, error analysis, and interpretation using computer-based tools for informed decision-making.

Syllabus

UNIT 1: Solution of Algebraic and Transcendental Equation: Errors and their Computation, Rate of convergence, Methods of finding roots: Iteration method, Bisection Method, Method of false position, Newton-Raphson method, Solution of system of linear equations: Matrix Decomposition methods, Jacobi method, Gauss-Seidel method.

UNIT 2: Finite Differences & Interpolation: Finite Differences, Difference tables Polynomial Interpolation: Newton's forward and backward formula Central Difference Formulae: Gauss forward and backward formula, Stirling's, Bessel's, Everett's formula. Interpolation with unequal intervals: Langrange's Interpolation, Newton Divided difference formula, Hermite's Interpolation.

UNIT 3: Numerical Integration and Differentiation: Introduction, Numerical differentiation Numerical Integration: Trapezoidal rule, Simpson's 1/3 and 3/8 rule, Boole's rule, Waddle's rule. Solution of differential Equations: Picard's Method, Euler's Method, Taylor's Method, Runge-Kutta Methods, Predictor Corrector Methods, Automatic Error Monitoring and Stability of solution.

UNIT 4: Statistical Computation: Frequency chart, Curve fitting by method of least squares, fitting of straight lines, polynomials, exponential curves etc, Data fitting with Cubic splines, Correlation and Regression Analysis: Linear and Nonlinear Regression.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Implement numerical methods like Bisection, Regula-Falsi, and Newton-Raphson for solving algebraic and transcendental equations.

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C02: Apply interpolation, numerical differentiation, and integration techniques to approximate mathematical functions and solve practical problems.

C03: Solve systems of linear equations and differential equations using numerical methods such as Gauss-Seidel, Euler, and Runge-Kutta.

C04: Use statistical techniques including correlation, regression, probability distributions for data analysis and interpretation.

Textbooks:

1. Peter V. O'Neil, Advanced Engineering Mathematics Thomson (Cengage) Learning, 2007.
2. Jain, Iyenger & Jain, Numerical Methods for Scientific and Engineering Computation, New Age International, New Delhi.
3. JN Kapur, Mathematical Statistics, S. Chand & company Ltd.
4. BS Grewal, Higher Engineering Mathematics, Khanna Publishers.
5. R.K. Jain & SRK Iyenger, Advanced Engineering Mathematics, Narosa Publication House.
6. Chandrika Prasad, Advanced Mathematics for Engineers, Prasad Mudralaya, 1996.
7. S. S. Sastry, Introductory Methods of Numerical Analysis, PHI Learning Pvt. Limited, New Delhi
8. E. Balagurusamy, Numerical Methods, Tata McGraw-Hill Publishing Company Limited, New Delhi
9. T. Veerajan & T. Ramchandrandran, Theory & Problems in Numerical Methods, TMH, New Delhi.

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Natural Language Processing

Course Code: AID309

Credit: 3+0

Course Objective: The objectives of this course are to provide a foundational understanding of natural language processing (NLP), enable students to process and analyse human language data using computational techniques, and develop proficiency in applying NLP methods and algorithms.

Syllabus

UNIT 1: Introduction to Natural Language Processing: Tokenization, Stopword Removal, Stemming and Lemmatization, Part-of-Speech (POS) Tagging, Named Entity Recognition (NER), Text Normalization, Language Models, Word Embeddings, Text Classification.

UNIT 2: Text Representation and Feature Engineering: Bag-of-Words (BoW) Model, TF-IDF Representation, Word2Vec Embeddings, GloVe Embeddings, Document Embeddings, Feature Selection Techniques, Feature Extraction Methods, Topic Modeling, Dimensionality Reduction.

Syntactic and Semantic Analysis: Syntax Analysis (Parsing), Constituency Parsing, Dependency Parsing, Semantic Role Labeling (SRL).

UNIT 3: Advanced NLP Techniques: Sentiment Analysis, Aspect-Based Sentiment Analysis, Coreference Resolution, Semantic Similarity. Sequence-to-Sequence Models, Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) Networks, Attention Mechanisms, Transformer Models, Pretrained Language Models (BERT, GPT), Transfer Learning in NLP, Neural Machine Translation, Text Summarization.

UNIT 4: Applications of Natural Language Processing: Named Entity Recognition in Healthcare, Sentiment Analysis in Social Media, Text Classification in Finance, Machine Translation in Multilingual Communication, Question Answering Systems, Chatbots and Virtual Assistants, Information Extraction from Text, Ethical Considerations in NLP

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: To explain theory underlying machine learning

CO2: To construct algorithms to learn linear and non-linear models

CO3: To implement data clustering algorithms

CO4: To construct algorithms to learn tree and rule-based models

CO5: To apply reinforcement learning techniques

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Textbooks:

1. "Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition" by Daniel Jurafsky and James H. Martin.
2. "Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit" by Steven Bird, Ewan Klein, and Edward Loper.
3. "Foundations of Statistical Natural Language Processing" by Christopher D. Manning and Hinrich Schütze.
4. "Deep Learning for Natural Language Processing" by Palash Goyal, Sumit Pandey, Karan Jain, and Karan Mittal.
5. "Applied Text Analysis with Python: Enabling Language-Aware Data Products with Machine Learning" by Benjamin Bengfort, Tony Ojeda, and Rebecca Bilbro.

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Computer Vision: Foundation and Applications

Course Code: AID310

Credit: 3+0

Course Objective: The objective of this course is to equip students with a comprehensive understanding of 2D image processing techniques and to develop their ability to extract and interpret meaningful information about objects in a scene through the study of projection geometry and 3D imaging concepts.

Syllabus

UNIT 1: Introduction: Computer Vision, Brief History. Image Formation: Geometric Primitives and Transformations, Photometric Image Formation, Digital Camera. Image Processing: Point Operators, Linear Filtering, Neighborhood Operators, Fourier Transform, Pyramids and Wavelet, Geometric Transforms, Global Optimization

UNIT 2: Feature Detection and Matching: Points and Patches, Edges, Lines. **Segmentation:** Active Contours: Snakes, Dynamic Snake and Condensation, Scissor, Level Sets, Split and Merge, Mean Shift and Mode Finding, **Feature Based Alignment:** 2D and 3D Feature Based Alignment, Pose Estimation, Geometric Intrinsic Calibration.

UNIT 3: Structure from Motion: Triangulation, Two Frame Structure from Motion, Factorization, Bundle Adjustment. **Dense Motion Estimation:** Translational Alignment, Parametric Motion, Spline Based Motion, Layered Motion. **Image Stitching:** Motion Models, Global Alignment, Composing.

UNIT 4: 3D Reconstruction: Surface Representation, Point based Representation, Volumetric Representation, Model based Reconstruction, Application: 3D Photography. **Image Based Rendering:** View Interpolation, Layered Depth Images, Video based Rendering. **Recognition:** Object Detection, Face Recognition, Context and Scene Understanding.

Course Outcomes (COs): After completion of this course, students will be able to:

CO1: Apply various techniques and operations for processing and enhancing 2D images.

CO2: Analyze image data to extract meaningful information about objects within a scene.

CO3: Interpret projection geometry to understand the formation and representation of 3D images.

CO4: Reconstruct and visualize 3D information from 2D images using appropriate methods.

CO5: Evaluate and apply image processing techniques for real-world applications in computer vision and imaging systems.

Textbooks:

1. R. Szeliski, Computer Vision: Algorithms and Applications, Springer.

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2. D. Forsyth and J. Ponce, Computer Vision- A Modern Approach, Prentice Hall.

Problem Solving Method in Artificial Intelligence

Course Code: AID311

Credit: 3+0

Course Objectives: This course develops problem-solving skills using AI techniques, covering fundamental concepts, algorithms, and tools for decision-making, optimization, and designing intelligent solutions for real-world applications.

Syllabus

UNIT 1: Introduction to AI Problem Solving: Introduction to Artificial Intelligence, Problem Solving using AI, Intelligent Agents and Environments, State Space Representation, Problem Formulation and Characteristics, Overview of AI Techniques

UNIT 2: Search Algorithm: Techniques like Breadth First Search (BFS), Depth First Search (DFS), Hill Climbing, A* Algorithm, and Constraint Satisfaction Problems (CSPs).

UNIT 3: Knowledge Representation: Importance of Knowledge Representation in AI, Predicate Logic, semantics Networks, frames and Scripts, reasoning and Inference mechanism.

UNIT 4: Planning and Expert Systems: AI Planning and Goal-based Agents, Planning Algorithms and Strategies, Introduction to Expert Systems, Components of Expert Systems, Applications of Expert Systems in real-world problems.

Course Outcome (COs): After completion of this course, students will be able to:

CO1: Interpret fundamental AI concepts and intelligent agent models.

CO2: Model real-world problems using state space and problem formulation techniques.

CO3: Implement and compare classical and heuristic search algorithms.

CO4: Apply knowledge representation schemes and reasoning mechanisms.

CO5: Design AI-based planning systems and goal-driven agents.

CO6: Analyze and develop expert systems for practical applications.

Textbook:

1. Artificial Intelligence: A Modern Approach, Authors: Stuart J. Russell & Peter Norvig
2. Artificial Intelligence: Structures and Strategies for Complex Problem Solving
Author: George F. Luger
3. Patrick Winston – *Artificial Intelligence*

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Mathematics for Data Science

Course Code: EMAT105

Credit: 3+0

Course objectives: This course introduces mathematical foundations for data science, including sets, relations, functions, linear systems, matrices, vector spaces, calculus, differential equations, and optimization for modern computing applications.

Syllabus

UNIT 1: Set Theory: Definition of sets, representation of sets, universal set, empty set, singleton set, finite and infinite set, equal set, subsets, proper subset, superset, power set, improper set, comparability of sets, union and intersection of sets, complement of sets, De- Morgan's law, disjoint sets, difference and symmetric difference, Venn diagram and its applications. Relations, equivalence relation, functions or mapping, domain, co domain and range of a function, inverse of function, composition of function.

UNIT 2: Differential Calculus: Sequences and series, Limits, continuity and Derivatives of functions, Differentiation rules, Chain rule, Implicit differentiation, Related rates, Optimization, Curve sketching, L'Hôpital's rule, Maxima and minima and saddle points, Applications of differentiation. Integral Calculus: Antiderivatives, Definite integrals, Integration techniques (substitution, integration by parts, trigonometric substitution), Improper integrals, Area under curves, Volume of revolution, Arc length, Surface area, Applications of integrals.

UNIT 3: Determinant and Matrices: Introduction to matrix, properties of matrix; evaluation of determinant, rank, minor and cofactors and Properties of determinant, different operations on matrices, Eigen values and Eigen vectors, System of linear equations, Solving by substitution, solving by elimination, solving with matrices, Gaussian elimination, Cramer's rule, homogeneous systems, non-homogeneous systems, consistency assessment..

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UNIT 4: Differentials Equation: Modelling with Differential Equations, Linear and Bernoulli's differential equations, Nonlinear differential equations, Linear differential equations of second and higher order with constant coefficients: Non- homogeneous term of the type $f(x) = e^{ax}, \sin ax, \cos ax, x^n, e^{ax} V$ and $x^n V$ - Method of variation of parameters, equations reducible to linear ODE with constant coefficients-Cauchy's Euler equation and Legendre's equation.

Course Outcome (COs): After completing this course, students will be able to:

CO1: Able to study the concepts of matrices and apply them in related engineering problems.

CO2: Extend the use of linear systems of equations, matrices and determinants, and vector spaces in the science of data.

CO3: Able to extend the concepts of integral calculus in finding area and volume.

CO4: Capable to use the features of Differential Calculus in optimization problems.

CO5: Skilled to solve ordinary differential equations in engineering problems.

CO6: Articulate the use of different optimization techniques for data analysis.

CO7: Illustrate analytical models for real-word scenarios.

Textbooks:

1. Grewal B.S., "Higher Engineering Mathematics", Khanna Publishers, New Delhi, 43rd Edition, 2014.
2. Bali N. P and Manish Goyal, "A Textbook of Engineering Mathematics", Eighth Edition, Laxmi Publications Pvt Ltd., 2011.
3. Chandrasekaran A, "A Text book of Engineering Mathematics I", Dhanam Publications, Chennai, 2017.
4. Srimantha Pal and Bhunia, S.C, "Engineering Mathematics" Oxford University Press, 2015
5. Weir, M.D and Joel Hass, Thomas' Calculus, 12th Edition, Pearson India, 2016.
6. Advanced Engineering Mathematics With Matlab, Third Edition, 2011 by CRC Press.

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Mathematical Foundation of Machine Learning

Course Code: EMAT106

Course Objective: Students will be able to learn the fundamentals of mathematical concepts used in the area of machine Learning. Students will learn the concept of probability, statistics and linear algebra for machine learning.

Syllabus

UNIT 1: Equation and Polynomial: Introduction to Machine Learning, Linear equation and solution, Intercepts and slopes, Exponentials, radicals and logarithms, Polynomials, Polynomial operations, Factorizations, Quadratic equation.

UNIT 2: Linear Algebra: Vector space, subspaces, linear dependence and independence of vectors, matrices, projection matrix, orthogonal matrix, idempotent matrix, partition matrix and their properties, quadratic forms, systems of linear equations and solutions, Matrix decomposition (LU, QR, SVD and Cholesky decomposition).

UNIT 3: Calculus & Optimization: Functions of one and several variables, Limits and continuity, Partial derivatives, Taylor series, Gradient, Jacobian, Hessian, Convex functions, Gradient descent algorithm, Optimization with one and multiple variable, Constrained and Unconstrained optimization.

UNIT 4: Statistics & Information Theory: Descriptive statistics, Probability, Random variables (discrete & continuous), Quantile Function, Probability distributions: Bernoulli, Binomial, Poisson, Normal distribution. Sampling and estimation, Bias and variance, Hypothesis testing, Maximum Likelihood Estimation (MLE), Maximum A Posteriori (MAP), Model Selection & Information theory: Entropy, Cross-entropy, KL divergence. AIC, BIC and HQIC.

Course Outcomes (CO): At the end of this course students will be able to:

CO1: To develop simple algorithms for arithmetic and logical problems.

CO2: To translate the algorithms to programs & execution (in C language).

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C03: To implement conditional branching, iteration and recursion.

C04: To decompose a problem into functions and synthesize a complete program using divide and conquer approach.

C05: To use arrays, pointers and structures to develop algorithms and programs.

Textbooks:

1. Christopher M. Bishop. 2006. Pattern Recognition and Machine Learning (Information Science and Statistics). Springer-Verlag New York, Inc., Secaucus, NJ, USA.
2. Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, 2020, Mathematics For Machine Learning. MIT-Press.